

Full Derivation Ledger for the Branch A Higgs Module

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Abstract

This companion ledger expands the derivation chain behind Paper 4. It is not an empirical paper, not a complete Standard Model derivation, and not a claim that the Higgs sector has been solved. Its purpose is to collect the working Branch A gate stack in one generated PDF: stabilizer normalization, localization forcing, quartic overlap, Yukawa product, bridge-trace normalization, v0.2 single-package parent action, absolute-scale gate, wall-cell embedding, and the remaining top/BRST/regulator blockers.

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1 Status Discipline

This ledger uses four statuses.

Status	Meaning
Derived in gate	Follows inside the stated Branch A assumptions or audit.
Theorem-candidate	Would become a theorem if the listed parent conditions are derived.
Open gate	Required before a physical claim can be made.
Forbidden overclaim	A claim explicitly not licensed by the current construction.

The guiding rule is simple: no quantity may be chosen from the observed Higgs vev, measured Higgs mass, or top mass and then presented as derived.

2 Global Dependency Chain

The current Higgs-route dependency chain is

$$3+2 \rightarrow T_Y \rightarrow \kappa_\tau^2 = \frac{3}{5} \rightarrow Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x \rightarrow \nu_i = \kappa_\tau^2 |Y_i| \rightarrow I_4(3/10) \rightarrow \mathcal{I}_{qHu} \rightarrow y_i^{\text{parent}} \rightarrow \Lambda_A \rightarrow m_t^{\text{candidate}}. \quad (1)$$

The first part is dimensionless and comparatively sharp. The last part is a scale/matching problem and remains open.

3 Branch A Stabilizer And Hypercharge Normalization

Start from the visible carrier

$$V \simeq \mathbb{C}^5 = C \oplus W, \quad \dim C = 3, \quad \dim W = 2. \quad (2)$$

The traceless two-block generator is

$$T_\Sigma = \frac{1}{\sqrt{60}} \text{diag}(2, 2, 2, -3, -3). \quad (3)$$

The stabilizer is $S(U(3) \times U(2))$. A stabilizer-preserving Abelian connection must be block-scalar:

$$T = \text{diag}(a, a, a, b, b), \quad 3a + 2b = 0. \quad (4)$$

Thus the traceless centralizer is one-dimensional. Choosing the conventional hypercharge orientation gives

$$Y = \text{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right), \quad T_Y = \sqrt{\frac{3}{5}} Y, \quad (5)$$

and $T_\Sigma = -T_Y$. The Branch A normalization is therefore

$$\kappa_\tau^2 = \frac{3}{5}. \quad (6)$$

This is not fitted to the Higgs quartic. It comes from canonical normalization of the unique traceless Abelian line in the $3 + 2$ stabilizer.

4 Spectral Origin Of The Normalization Gate

The next upgrade is to derive the same coefficient from the compact spectral geometry rather than from a normalization convention. In the v0.1 compact geometry, the target is

$$\kappa_\tau^2 = \frac{\text{Tr}_{\mathcal{F}}(T_Y^2)}{\text{Tr}_{\mathcal{F}}(Y^2)} = \frac{3}{5}, \quad (7)$$

where the finite fiber $\mathcal{F} = \mathbb{C}^3 \oplus \mathbb{C}^2$ and trace functional are selected by the compact domain. A more intrinsic version would express the same value as a heat-kernel or index residue:

$$\kappa_\tau^2 \sim \frac{\text{Res}_{s=0} \text{Tr}(T_Y^2 |D_\tau|^{-s})}{\text{Res}_{s=0} \text{Tr}(Y^2 |D_\tau|^{-s})}. \quad (8)$$

The proof obligation is therefore:

1. the compact spectrum must select the trace functional;
2. the protected hypercharge line must be an eigenline or index residue of the compact operator;
3. the residue ratio must be stable under allowed regulator choices;
4. no alternative Abelian line may give a lower selection penalty.

If this closes, the factor $3/5$ becomes a spectral number of the compact geometry. Combined with the charge-to-exponent theorem target, this would turn $\nu_D = 3/10$ into a genuine geometric/spectral consequence rather than a conditional Branch A input.

5 Unoriented Line And Two-Vacuum Quotient

The Branch A Abelian direction is treated as an internal line rather than as a signed observable vector:

$$L_Y = \text{span}(T_Y). \quad (9)$$

If the regular parent datum is the line or projector, then

$$\text{span}(T_Y) = \text{span}(-T_Y), \quad P_Y = P_{-Y}, \quad \Sigma \sim -\Sigma. \quad (10)$$

For $\Sigma = fT_Y$, sign-even invariants satisfy

$$\text{Tr}(\Sigma^2) = \frac{1}{2}f^2, \quad \text{Tr}(\Sigma'^2) = \frac{1}{2}(f')^2. \quad (11)$$

If the regular branch fixes $2\text{Tr}(\Sigma^2) = 1$, then $f^2 = 1$ and the two oriented representatives are $f = \pm 1$. This gives the two-vacuum logic conditional on the parent deriving the unoriented-line/projector quotient.

6 Minimal Wall Profile

The leading local two-vacuum wall is modeled by the BPS equation

$$f'(x) = 1 - f(x)^2, \quad (12)$$

whose centered solution is

$$f(x) = \tanh x. \quad (13)$$

Equivalently, it is generated by

$$V(f) = \frac{1}{2}(1 - f^2)^2. \quad (14)$$

The wall is not selected to fit the Higgs quartic. The open parent-level task is to derive this minimal BPS functional and forbid leading deformations that change the localization rule.

7 Localization Derivation

Given the selected hypercharge wall line, the local first-order operator is

$$Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x. \quad (15)$$

The zero-mode equation gives

$$Q_i h_i = 0, \quad (16)$$

$$\partial_x h_i = -\kappa_\tau^2 Y_i \tanh x h_i, \quad (17)$$

$$h_i(x) \propto \exp \left[- \int^x \kappa_\tau^2 Y_i \tanh u du \right] = \text{sech}^{\kappa_\tau^2 Y_i} x. \quad (18)$$

Normalizability selects the chirality/sign, so

$$\nu_i = \kappa_\tau^2 |Y_i|. \quad (19)$$

For the Higgs doublet,

$$\nu_H = \frac{3}{5} \left| \frac{1}{2} \right| = \frac{3}{10}. \quad (20)$$

This is the current natural-origin derivation. It becomes a proof only when the parent action derives the wall line, BPS wall, same-domain Hessian, and chiral admissibility.

8 Charge-To-Exponent Theorem Target

The value $\nu_H = 3/10$ has two ingredients. The first is already algebraic: canonical Branch A normalization gives

$$\kappa_\tau^2 = \frac{3}{5}, \quad |Y_D| = \frac{1}{2}. \quad (21)$$

The second is still the hard gate: the compact geometry must show that the wall connection seen by a component with hypercharge Y_i is exactly

$$A_{\tau,i}(x) = \kappa_\tau^2 Y_i f(x) \quad (22)$$

on the self-adjoint wall domain. If this is derived, the first-order local operator is forced:

$$Q_i = \partial_x + A_{\tau,i}(x), \quad Q_i h_i = 0. \quad (23)$$

For the Bogomolny wall $f(x) = \tanh x$,

$$h_i(x) \propto \exp \left[-\kappa_\tau^2 Y_i \int^x \tanh u \, du \right] = \text{sech}^{\kappa_\tau^2 Y_i} x, \quad (24)$$

and normalizability gives

$$\nu_i = \kappa_\tau^2 |Y_i|. \quad (25)$$

Therefore the derivation problem is now sharply localized. One must prove that the compact spectral geometry selects a universal hypercharge-line connection coefficient $\kappa_\tau^2 = 3/5$ and forbids extra component-dependent connection terms:

$$A_{\tau,i}(x) \neq \kappa_\tau^2 Y_i f(x) + \delta A_i(x) \quad (26)$$

unless δA_i is Q-exact, quotient-null, or massive/leakage-suppressed. This is the precise theorem that would turn the current 3/10 input into a geometric/spectral consequence.

9 F1–F8 Forcing Theorem Candidate

The stronger theorem-candidate says the localization rule is forced if the parent action derives F1–F8.

Gate	Requirement	Consequence
F1	quotient-visible trace removal	identity direction cannot become a wall channel
F2	two protected visible blocks	exactly two low-energy visible modules
F3	stabilizer-preserving wall	wall is block-scalar
F4	one traceless Abelian wall line	no extra Abelian wall freedom
F5	universal trace pairing	fixes $\kappa_\tau^2 = 3/5$
F6	unoriented two-vacuum quotient	gives the $\pm$ wall representatives
F7	minimal BPS factorization	gives leading $f = \tanh x$
F8	same-domain Hessian/Q factorization	forbids importing Q_i by hand

Under these conditions the operator Q_i and the exponent ν_i follow at leading order. Still open: deriving F1–F8 from the final parent action and stationary background.

10 Minimal Forcing Parent-Action Candidate

The current action-level scaffold behind F1–F8 contains:

- trace projection Π_{tr} to remove the identity direction;
- a stabilizer-stratified wall sector;
- positive transverse/off-block gaps;
- an unoriented-line quotient sector;
- a minimal BPS wall term;
- a same-domain Hessian/Q admissibility term.

It forces the trace quotient, trace pairing, leading BPS wall, and local localization consequence at gate level. It does not yet fully force the two protected visible modules, the deep origin of the unoriented-line quotient, or continuum QFT closure.

11 Quartic Overlap Calculation

For

$$h_\nu(x) = N_\nu \text{sech}^\nu x, \quad \int h_\nu^2 dx = 1, \quad (27)$$

the normalization is

$$N_\nu^2 = \frac{\Gamma(\nu + 1/2)}{\sqrt{\pi}\Gamma(\nu)}. \quad (28)$$

The quartic overlap is

$$I_4(\nu) = \frac{\Gamma(\nu + 1/2)^2 \Gamma(2\nu)}{\sqrt{\pi}\Gamma(\nu)^2 \Gamma(2\nu + 1/2)}. \quad (29)$$

At $\nu_H = 3/10$,

$$I_4(3/10) = 0.133756368895144. \quad (30)$$

For $\lambda_H \simeq 0.129$,

$$\lambda_\tau \simeq 0.964440057. \quad (31)$$

The safe claim is order-one parent quartic, not a derived canonical quartic.

12 Sensitivity Audit

The generator also writes a sensitivity audit. In the moderate window $0.25 \leq \nu \leq 0.35$, the required parent quartic remains order one. This reduces numerology risk: the mechanism is not a delta-function coincidence at $\nu = 0.3$. The audit still does not prove the localization rule.

13 Yukawa Product Cohomology Gate

The up-type slot is treated as a parent cohomology product, not a post-hoc operator insertion:

$$[q_L] [H] [u_R^c]. \quad (32)$$

It is admissible only if it is neutral, bridge-closed, Q-closed modulo Q-exact terms, and defined on the same Branch A domain as the wall/Hessian package. The gate does not derive family hierarchy or the measured top mass.

14 Yukawa Overlap

The fixed exponents are

$$\nu_q = \frac{1}{10}, \quad \nu_H = \frac{3}{10}, \quad \nu_u = \frac{2}{5}. \quad (33)$$

With

$$I(a) = \int_{\mathbb{R}} \text{sech}^a x \, dx = \sqrt{\pi} \frac{\Gamma(a/2)}{\Gamma((1+a)/2)}, \quad (34)$$

the normalized overlap is

$$\mathcal{I}_{qHu} = N_{\nu_q} N_{\nu_H} N_{\nu_u} I(\nu_q + \nu_H + \nu_u) = 0.267097842200176. \quad (35)$$

This is reproducible and endpoint-free.

15 g_t Bridge-Trace Normalization

The current normalization theorem-candidate sums over the full 3×2 shared bridge and multiplies by the canonical wall coupling:

$$g_t = (3 \times 2) \kappa_\tau^2 = 6 \cdot \frac{3}{5} = \frac{18}{5}. \quad (36)$$

Therefore

$$y_t^{\text{parent}} = g_t \mathcal{I}_{qHu} = 0.961552231920634. \quad (37)$$

This is close to an order-one top Yukawa, but it is not a top-mass prediction. The measured value still requires v_{eff} , R_t , family selection, and QFT matching.

16 v_{eff} From Stabilizer Gap

The current electroweak-scale gate proposes

$$v_{\text{eff}} = \frac{M_{\text{stab}}}{g_{\text{vis}}}. \quad (38)$$

Here M_{stab} must be the parent-derived positive Hessian/stabilizer gap protecting the Higgs-like residue, and g_{vis} must be a universal visible gauge normalization. This is better than inserting 246 GeV, but it is not yet a derivation of the measured vev.

17 Running And Matching

The admissible matching factor is

$$R_t = R_{\text{match}} R_{\text{run}}, \quad R_{\text{run}} = \exp \left[- \int \gamma_t(\mu) d \ln \mu \right]. \quad (39)$$

This is a shape requirement only. γ_t , R_{match} , regulator, and matching scale must come from the same QFT closure before a physical top mass can be claimed.

18 Physical Matching Admissibility Map

The physical matching problem is now a forward map, not a backsolve:

$$S_{\text{parent}} \longrightarrow (\epsilon_\tau, C_A, \mu_\tau(\mathcal{K}_\tau), N_\tau, \hat{E}_{\text{wall}}) \longrightarrow \Lambda_A \longrightarrow (v_{\text{eff}}, y_t^{\text{phys}}, m_H, m_t, \Delta g/g). \quad (40)$$

The forbidden reverse map is

$$(v_{\text{obs}}, m_H^{\text{obs}}, m_t^{\text{obs}}) \not\Rightarrow (\epsilon_\tau, C_A, R_t). \quad (41)$$

Thus physical matching is admissible only if the absolute scale, running, threshold correction, and visible gauge normalization are fixed upstream of the measured Higgs and top endpoints. A successful matching calculation may compare to the measured vev, Higgs mass, top mass, and collider coupling constraints, but it may not use those numbers to define the parent scale or the top matching factor.

19 Numerical Endpoint Matching Protocol

The endpoint comparison can also be made falsifiable before it is executed. Once the parent calculation outputs

$$\Theta_{\text{pred}} = (\epsilon_\tau, C_A, \mu_\tau, N_\tau, \hat{E}_{\text{wall}}, R_t^{\text{fin}}, Z_H, Z_t), \quad (42)$$

the matching code must freeze these numbers and compute

$$\mathcal{M}(\Theta_{\text{pred}}) = (v_{\text{eff}}, m_H, m_t, \Delta g_H, \Delta g_t). \quad (43)$$

Only then may the result be compared with the observed endpoint vector

$$\mathcal{E}_{\text{obs}} = (v_{\text{obs}}, m_H^{\text{obs}}, m_t^{\text{obs}}, \Delta g_H^{\text{obs}}, \Delta g_t^{\text{obs}}). \quad (44)$$

The pass/fail statistic should be declared before looking at the residuals, for example

$$\chi_{\text{end}}^2 = (\mathcal{M}(\Theta_{\text{pred}}) - \mathcal{E}_{\text{obs}})^T \Sigma_{\text{obs}}^{-1} (\mathcal{M}(\Theta_{\text{pred}}) - \mathcal{E}_{\text{obs}}). \quad (45)$$

This does not perform numerical endpoint matching. It defines the allowed order of operations: derive, freeze, run, compare. Any workflow that adjusts ϵ_τ , C_A , R_t^{fin} , or Z_H after seeing the endpoint residuals is a failed matching protocol, even if the final numbers look good.

20 Dimensional Obstruction

The Branch A action currently fixes dimensionless data. A dimensionless action cannot output an absolute GeV scale. Therefore

$$\Lambda_A = s_A \hat{E}_{\text{wall}} \quad (46)$$

requires one remaining absolute unit s_A .

21 s_A And Compact Tau-Cell Unit

The most native route is

$$s_A = C_A E_\tau, \quad E_\tau = \epsilon_\tau \mu_\tau(\mathcal{K}_\tau) N_\tau. \quad (47)$$

This route is admissible only if the compact cell, its measure, the count/index density, and ϵ_τ are selected before Higgs/top comparison.

22 Compact Tau Geometry Target

The compact cell must become a concrete object rather than a name. The minimal target data are

$$(\mathcal{K}_\tau, \mathcal{A}_\tau, \mathcal{H}_\tau, D_\tau, \mu_\tau), \quad (48)$$

where $\mathcal{A}_\tau$ is an internal algebra, $\mathcal{H}_\tau$ is the domain, D_τ is the spectral operator, and μ_τ is the trace or measure. A useful candidate must produce the Branch A wall as a defect, quotient, or induced Hessian sector of this same package:

$$\mathcal{W}_A = \text{Defect}(\mathcal{K}_\tau, D_\tau, T_Y) \quad \text{or} \quad \mathcal{W}_A = \mathcal{K}_\tau / \mathcal{H}_A. \quad (49)$$

The decisive check is whether the low-lying spectral/Hessian data of $\mathcal{K}_\tau$ select 3+2, the hypercharge line, a two-vacuum kink, the same-domain Q package, and a finite determinant regulator without inserting those choices by hand.

23 C0 Finite-Fiber Wall Spectral Cell

The first explicit compact-geometry candidate uses a compact local wall cell with finite internal fiber:

$$\mathcal{H}_\tau = L^2(\mathcal{K}_\tau) \otimes \mathbb{C}^5 \otimes \mathbb{C}_{\text{wall}}^2, \quad \mathbb{C}^5 = C \oplus W, \quad \dim C = 3, \quad \dim W = 2. \quad (50)$$

Preserving this two-block fiber gives $S(U(3) \times U(2))$. The traceless block-scalar centralizer is one-dimensional:

$$T = \text{diag}(a, a, a, b, b), \quad 3a + 2b = 0, \quad (51)$$

which is the hypercharge line after canonical normalization.

The regular Abelian datum is taken to be the unoriented projector

$$P_Y = \frac{T_Y \otimes T_Y}{\text{Tr}(T_Y^2)}, \quad (52)$$

so T_Y and $-T_Y$ define the same parent line. The quotient-visible wall field $\Sigma = f T_Y$ then has the minimal BPS functional

$$E[f] = \int \left[\frac{1}{2} (f')^2 + \frac{1}{2} (1 - f^2)^2 \right] dx, \quad (53)$$

whose centered solution is $f(x) = \tanh x$. The same wall-Hessian factor gives

$$Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x, \quad \nu_i = \kappa_\tau^2 |Y_i|. \quad (54)$$

Because $\mathcal{K}_\tau$ is compact before the local-wall limit, the spectrum of the parent D_τ is discrete and can in principle define the same heat-kernel or zeta regulator used by the determinant gate:

$$\zeta_{D_\tau}(s) = \sum_{\lambda_n > 0} \lambda_n^{-s}. \quad (55)$$

This is the first worked compact tau geometry route. Its remaining blocker is that a final S_{parent} must select the two-block cell and its wall functional rather than inheriting them as useful choices.

24 Adversarial Audit Of C0

C0 survives only as a candidate. The attack map is:

- the $C^3 \oplus C^2$ fiber is still inserted; after insertion the stabilizer and hypercharge line follow, but parent selection remains open;
- the minimal wall functional is still chosen; after choice it gives $f(x) = \tanh x$, but the final action must force this functional or an equivalent BPS system;
- the line-projector quotient is assumed; after assumption it gives $\Sigma \sim -\Sigma$, but the parent regularity principle must choose projector data over signed-vector data;
- the compact spectrum gives a natural regulator route, but no physical top determinant, anomaly closure, or Ward identity has yet been computed;
- ϵ_τ remains a scale gate, not a derived number.

The C0 result is therefore not proof. It is a useful failure-localization device: the next C1 step must derive the two-block fiber and wall functional from a parent extremal, index, or stability principle.

25 C1 And C2 Forcing Routes

C1 addresses the two inserted C0 structures. The 3+2 fiber is no longer treated as a free five-dimensional choice if the parent action requires exactly two protected visible roles: a primitive closure role and a primitive pairing role. The minimal dimensions are then

$$\dim C = 3, \quad \dim W = 2, \quad (56)$$

because the first nontrivial oriented closure tensor is ϵ_3 and the first primitive pairing tensor is ϵ_2 . The BPS wall is no longer a free potential if the parent line datum is unoriented, local, reflection-even, finite-tension, and in the unit two-vacuum sector. The leading Bogomolny completion is

$$E[f] = \int \frac{1}{2} (f' - (1 - f^2))^2 dx + \left[f - \frac{1}{3} f^3 \right]_{-\infty}^{+\infty}, \quad (57)$$

so the minimizer satisfies $f' = 1 - f^2$ and therefore $f(x) = \tanh x$.

C2 addresses the remaining C0 attacks. Parent sign-gauge regularity would force the unoriented projector:

$$T_Y \mapsto -T_Y, \quad \mathcal{A}_{\text{phys}} = \mathcal{A}_\tau^{\mathbb{Z}_2}, \quad P_Y = \frac{T_Y \otimes T_Y}{\text{Tr}(T_Y^2)}. \quad (58)$$

Same-domain QFT closure would make the top-sensitive determinant a zeta regularized shared-domain object:

$$\Delta_{\text{top}}(s) = \frac{\det_\zeta(Q_t^\dagger Q_t + \delta_\star W_t)}{\det_\zeta(Q_t Q_t^\dagger)}. \quad (59)$$

Finally, the absolute scale would have to be computed from compact-cell spectral density rather than endpoint fitting:

$$\epsilon_\tau = \frac{1}{\mu_\tau(\mathcal{K}_\tau)} \text{Tr}_{\mathcal{H}_\tau} (\chi(D_\tau^2/\Lambda_\tau^2) D_\tau^2), \quad (60)$$

with χ fixed before Higgs/top comparison. These are still theorem-candidate routes. They are stronger than C0 because the failure modes are now explicit and calculational.

26 C3 Explicit Geometry Candidate

The C3 candidate declares the compact spectral geometry explicitly:

$$\mathcal{K}_\tau = S^1/\mathbb{Z}_2, \quad \mathcal{H}_\tau = L^2(\mathcal{K}_\tau) \otimes (\mathbb{C}^3 \oplus \mathbb{C}^2) \otimes \mathbb{C}_{\text{wall}}^2, \quad (61)$$

$$\mathcal{A}_\tau = (C^\infty(\mathcal{K}_\tau) \otimes (M_3(\mathbb{C}) \oplus M_2(\mathbb{C})))^{\mathbb{Z}_2}. \quad (62)$$

The determinant-one unitary sector gives $S(U(3) \times U(2))$, and the traceless block-scalar centralizer gives the hypercharge line. The local wall operator is

$$D_\tau = \sigma_1 \partial_x + \sigma_2 \kappa_\tau^2 T_Y f(x) + D_{\text{gap}}, \quad (63)$$

with positive D_{gap} on leakage modes. The leading local even finite-tension wall action on the orbifold line is

$$E[f] = \int \left[\frac{1}{2} (f')^2 + \frac{1}{2} (1 - f^2)^2 \right] dx, \quad (64)$$

so the large-cell local solution is $f(x) = \tanh x$. Component restriction gives $Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x$ and $\nu_i = \kappa_\tau^2 |Y_i|$.

C3 is the first explicit $(\mathcal{K}_\tau, \mathcal{A}_\tau, \mathcal{H}_\tau, D_\tau, \mu_\tau)$ candidate. It realizes the C0–C2 architecture in one object, but it is still not a proof until the final S_{parent} selects this orbifold base, two-role fiber, even algebra, and wall action.

27 C3.1 Compact Geometry v0.1 Computation Target

The v0.1 compact geometry is not only a mnemonic for the desired result; it defines a specific calculation. Choose a self-adjoint domain $\mathcal{D}(D_\tau)$ on $S^1/\mathbb{Z}_2$ such that the orbifold parity preserves the physical projector line while exchanging the two oriented representatives:

$$\Pi P_Y \Pi^{-1} = P_Y, \quad T_Y \sim -T_Y. \quad (65)$$

The zero-mode/index gate is then

$$\text{ind}(Q_i) = \dim \ker Q_i - \dim \ker Q_i^\dagger, \quad Q_i = \partial_x + \kappa_\tau^2 Y_i f(x). \quad (66)$$

For the local kink limit, the candidate gives the normalizable branch

$$h_i(x) = N_i \text{sech}^{\kappa_\tau^2 |Y_i|} x. \quad (67)$$

The compact theorem target is stronger: prove the same zero-mode residue and positive-spectrum pairing on the self-adjoint orbifold domain, not merely in the infinite-line local limit.

28 C3.1 Master Compact Spectral Theorem Statement

The local pieces can be condensed into the theorem that would change the status of the framework. Let $Q : \mathcal{D}(Q) \subset \mathcal{H}_- \rightarrow \mathcal{H}_+$ be the parent-selected wall operator on the compact orbifold domain, and set

$$H_- = Q^\dagger Q, \quad H_+ = Q Q^\dagger. \quad (68)$$

If the final parent action proves the following four properties,

1. Q is closed, densely defined, and Fredholm on a compact self-adjoint orbifold domain;
2. $H_{\pm}$ are non-negative self-adjoint operators with compact resolvent;
3. the anomaly/Ward domain is the same domain used by Q , $H_{\pm}$, and $\det_{\zeta}$;
4. the boundary form vanishes without endpoint-tuned spectator fields,

then the theorem-level consequences are fixed:

$$\text{Spec}_+(H_-) = \text{Spec}_+(H_+) \quad \text{with multiplicity,} \quad (69)$$

$$\text{ind}(Q) = \dim \ker Q - \dim \ker Q^{\dagger}, \quad (70)$$

$$\log \frac{\det_{\zeta} H_-^{\text{pos}}}{\det_{\zeta} H_+^{\text{pos}}} = 0. \quad (71)$$

All surviving finite terms must then belong to the explicit residue list

$$\log \Delta = \log \Delta_{\text{index}} + \log \Delta_{\text{anomaly}} + \log \Delta_{\text{boundary}} + \log \Delta_{\text{matching}}. \quad (72)$$

This is the real breakthrough theorem. C3.1b proves the abstract pairing lemma, C3.1c gives one explicit Fredholm relative-orbifold domain, C3.1d gives the compact self-adjoint spectrum consequence, and C3.1e checks compact resolvent for the relative-orbifold realization. What is still missing is the parent-selection and anomaly-compatible domain closure that would make this theorem physical rather than merely available.

29 C3.1a Orbifold Q-Domain And Index Theorem Target

The missing compact theorem can be stated as a boundary-domain problem. Let $I = [0, L]$ be the fundamental interval of $S^1/\mathbb{Z}_2$ and let

$$Q_{\nu} = \partial_x + \nu f(x), \quad Q_{\nu}^{\dagger} = -\partial_x + \nu f(x) \quad (73)$$

act between two parity sectors $\mathcal{H}_-$ and $\mathcal{H}_+$. Integration by parts gives the boundary form

$$\langle Q_{\nu} \psi, \phi \rangle_+ - \langle \psi, Q_{\nu}^{\dagger} \phi \rangle_- = [\psi^*(x) \phi(x)]_0^L. \quad (74)$$

Therefore a Q-compatible orbifold domain must impose boundary conditions for which this boundary form vanishes and for which

$$Q_{\nu} : \mathcal{D}_- \rightarrow \mathcal{D}_+, \quad Q_{\nu}^{\dagger} : \mathcal{D}_+ \rightarrow \mathcal{D}_-. \quad (75)$$

The theorem target is then:

1. $H_- = Q_{\nu}^{\dagger} Q_{\nu}$ and $H_+ = Q_{\nu} Q_{\nu}^{\dagger}$ are self-adjoint on the induced domains;
2. every positive H_- eigenmode is mapped by Q_{ν} to a positive H_+ eigenmode with the same eigenvalue, and conversely by $Q_{\nu}^{\dagger}$;
3. the unpaired residue is exactly

$$\text{ind}(Q_{\nu}) = \dim \ker Q_{\nu} - \dim \ker Q_{\nu}^{\dagger}; \quad (76)$$

4. for the Branch A wall sector this index is stable under compact orbifold regularization and equals the protected zero-mode count.

This is the real compact-index gate. The Q-first finite-dimensional demo proves the algebraic pairing mechanism once such a domain exists; it does not itself prove that the physical orbifold boundary conditions select the domain.

30 C3.1b Proof Of The Q-Pairing Lemma

The abstract pairing statement can be proved independently of the detailed Branch A physics. Let $Q : \mathcal{D}(Q) \subset \mathcal{H}_- \rightarrow \mathcal{H}_+$ be a closed densely defined operator with adjoint $Q^\dagger$, and define

$$H_- = Q^\dagger Q, \quad H_+ = QQ^\dagger. \quad (77)$$

Assume the boundary conditions make the integration-by-parts boundary form vanish, so that these are the adjoint operators on the stated domains. Then H_- and H_+ are non-negative self-adjoint operators on their induced domains. If

$$H_- \psi = \lambda \psi, \quad \lambda > 0, \quad (78)$$

then $Q\psi \neq 0$, because $Q\psi = 0$ would imply $\lambda \|\psi\|^2 = \langle \psi, H_- \psi \rangle = \|Q\psi\|^2 = 0$. Moreover,

$$H_+(Q\psi) = QQ^\dagger Q\psi = QH_- \psi = \lambda Q\psi. \quad (79)$$

Thus Q maps every positive H_- eigenvector to a positive H_+ eigenvector with the same eigenvalue. Conversely, if $H_+ \phi = \lambda \phi$ with $\lambda > 0$, then $Q^\dagger \phi \neq 0$ and

$$H_-(Q^\dagger \phi) = Q^\dagger QQ^\dagger \phi = Q^\dagger H_+ \phi = \lambda Q^\dagger \phi. \quad (80)$$

The maps are inverse up to the scalar λ :

$$Q^\dagger Q\psi = \lambda \psi, \quad QQ^\dagger \phi = \lambda \phi. \quad (81)$$

Therefore the positive eigenspaces are isomorphic. The only possible unpaired states lie at $\lambda = 0$, where

$$\ker H_- = \ker Q, \quad \ker H_+ = \ker Q^\dagger. \quad (82)$$

If Q is Fredholm on the compact domain, the unpaired residue is

$$\text{ind}(Q) = \dim \ker Q - \dim \ker Q^\dagger. \quad (83)$$

This proves the compact-index algebraic core: once a physical orbifold domain really supplies such a closed Fredholm Q , the positive spectra pair and only the index/anomaly/boundary/matching residues can remain. What is still open is the physical selection of that domain by the final parent action.

31 C3.1c Explicit Relative-Orbifold Domain Proof

One explicit compact-domain realization can now be proved. Let

$$\mathcal{H}_- = L^2([0, L]), \quad \mathcal{H}_+ = L^2([0, L]) \quad (84)$$

and choose the orbifold-relative domain pair

$$\mathcal{D}(Q_\nu) = H^1([0, L]), \quad \mathcal{D}(Q_\nu^\dagger) = H_0^1([0, L]). \quad (85)$$

Here H_0^1 means that the field vanishes at the two orbifold fixed points. This is the relative/Dirichlet parity sector, while H^1 is the maximal partner sector. The boundary form vanishes because

$$\phi(0) = \phi(L) = 0 \implies [\psi^* \phi]_0^L = 0 \quad (86)$$

for all $\psi \in H^1$ and $\phi \in H_0^1$. Therefore the formal adjoint of $Q_\nu = \partial_x + \nu f(x)$ on H^1 is exactly $Q_\nu^\dagger = -\partial_x + \nu f(x)$ on H_0^1 .

The operator $Q_\nu : H^1 \rightarrow L^2$ is closed and Fredholm on the compact interval: it is a first-order elliptic operator with compact Sobolev embedding $H^1 \hookrightarrow L^2$ and finite-dimensional kernel/cokernel. Its kernel is the solution space of

$$\partial_x \psi + \nu f(x) \psi = 0, \quad \psi(x) = C \exp \left[-\nu \int_0^x f(u) du \right]. \quad (87)$$

This solution lies in $H^1([0, L])$ for smooth bounded f , so

$$\dim \ker Q_\nu = 1. \quad (88)$$

The adjoint kernel obeys

$$-\partial_x \phi + \nu f(x) \phi = 0, \quad \phi \in H_0^1([0, L]). \quad (89)$$

Its nonzero solutions cannot vanish at both endpoints unless the integration constant is zero. Hence

$$\dim \ker Q_\nu^\dagger = 0. \quad (90)$$

Thus the relative-orbifold domain has

$$\text{ind}(Q_\nu) = 1. \quad (91)$$

By the Q-pairing lemma, the positive spectra of $Q_\nu^\dagger Q_\nu$ and $Q_\nu Q_\nu^\dagger$ are paired, and the only unpaired compact residue is the single protected zero mode. This proves an explicit compact-domain theorem for the v0.1 relative-orbifold choice.

The remaining physics question is no longer whether such a compact Fredholm Q exists. It does. The remaining question is whether the final parent action uniquely selects this relative-orbifold domain rather than another self-adjoint extension.

32 C3.1c' Q-Compatible Orbifold Extension Classification

The self-adjoint extension problem can be reduced to boundary data. For $Q_\nu = \partial_x + \nu f(x)$ on $[0, L]$, the first-order boundary pairing is

$$B(\psi, \phi) = [\psi^* \phi]_0^L = \psi(L)^* \phi(L) - \psi(0)^* \phi(0). \quad (92)$$

A Q-compatible extension must choose boundary subspaces $\mathcal{B}_- \subset \mathbb{C}^2$ for $\psi|_{0,L}$ and $\mathcal{B}_+ \subset \mathbb{C}^2$ for $\phi|_{0,L}$ such that

$$B(\psi, \phi) = 0 \quad \text{for all } \psi|_{0,L} \in \mathcal{B}_-, \phi|_{0,L} \in \mathcal{B}_+. \quad (93)$$

Thus $\mathcal{B}_+$ must be contained in the boundary-form annihilator of $\mathcal{B}_-$. The relative-orbifold choice is the extremal pair

$$\mathcal{B}_- = \mathbb{C}^2, \quad \mathcal{B}_+ = \{0\}. \quad (94)$$

It is maximal on the minus side and Dirichlet on the plus side, hence it removes the boundary form without adding a tunable phase, endpoint mixing matrix, or Robin angle. The induced second-order conditions are

$$\phi|_{0,L} = 0, \quad Q_\nu \psi|_{0,L} = 0, \quad (95)$$

which are exactly the Dirichlet/Robin pair used above.

Other self-adjoint extensions are mathematically possible. For example, phase-periodic or mixed endpoint conditions can make second-order operators self-adjoint. They are not Q-compatible minimal survivors unless they also preserve the first-order pairing, the index residue, orbifold exchange symmetry, and the no-free-endpoint-parameter rule. In the C4 lexicographic ordering, the relative-orbifold pair

is therefore the first minimal survivor: it has no endpoint angle, no endpoint mixing, no field-specific boundary coefficient, and keeps the single protected index residue.

This is not yet a complete classification of all self-adjoint extensions. It is a classification of the Q-compatible minimal class relevant to the compact spectral theorem. A lower-complexity Q-compatible extension with the same index, no endpoint parameter, and anomaly-compatible domain closure would falsify this selection claim.

33 C3.1d Conditional Compact Self-Adjoint Spectrum Theorem

The preceding results give a conditional compact spectral theorem needed by the determinant gate. Assume:

1. $Q : \mathcal{D}(Q) \subset \mathcal{H}_- \rightarrow \mathcal{H}_+$ is closed, densely defined, and Fredholm;
2. the compact orbifold domain makes the boundary form vanish, so $Q^\dagger$ is the true adjoint on $\mathcal{D}(Q^\dagger)$;
3. $H_- = Q^\dagger Q$ and $H_+ = QQ^\dagger$ have compact resolvent.

Then H_- and H_+ are non-negative self-adjoint operators with discrete spectrum of finite multiplicity:

$$0 \leq \lambda_0^\pm \leq \lambda_1^\pm \leq \dots, \quad \lambda_n^\pm \rightarrow \infty. \quad (96)$$

By the Q-pairing lemma, the positive spectral subspaces are isomorphic:

$$\text{Spec}_+(H_-) = \text{Spec}_+(H_+) \quad (97)$$

with multiplicities. The only unpaired part is

$$\ker H_- - \ker H_+ = \ker Q - \ker Q^\dagger, \quad (98)$$

whose signed dimension is $\text{ind}(Q)$.

Define positive-spectrum zeta functions by

$$\zeta_-^{\text{pos}}(s) = \sum_{\lambda_n^- > 0} (\lambda_n^-)^{-s}, \quad \zeta_+^{\text{pos}}(s) = \sum_{\lambda_n^+ > 0} (\lambda_n^+)^{-s}. \quad (99)$$

Because the positive spectra agree with multiplicity,

$$\zeta_-^{\text{pos}}(s) = \zeta_+^{\text{pos}}(s) \quad (100)$$

where both sides are defined, and by analytic continuation wherever the zeta regularization is admissible. Therefore the positive determinant ratio cancels:

$$\log \frac{\det_\zeta H_-^{\text{pos}}}{\det_\zeta H_+^{\text{pos}}} = -\zeta_-^{\text{pos}}{}'(0) + \zeta_+^{\text{pos}}{}'(0) = 0. \quad (101)$$

Thus the determinant gate is reduced to residues:

$$\log \Delta = \log \Delta_{\text{index}} + \log \Delta_{\text{anomaly}} + \log \Delta_{\text{boundary}} + \log \Delta_{\text{matching}}. \quad (102)$$

This establishes the compact self-adjoint spectral proposition inside the relative-orbifold v0.1 domain. In that domain, the index residue is one. The remaining nontrivial physics gates are anomaly compatibility, regulator choice, boundary residue classification, and parent-action selection of this domain.

34 C3.1d' Protected Index Residue Theorem

For the Q-compatible relative-orbifold domain of C3.1c and C3.1c', the kernel structure can be stated as a protected index theorem. The minus-sector kernel is generated by

$$\psi_0(x) = C \exp \left[-\nu \int_0^x f(u) du \right], \quad (103)$$

which lies in $H^1([0, L])$ for smooth bounded f . The plus-sector adjoint kernel satisfies

$$Q_\nu^\dagger \phi = 0, \quad \phi(0) = \phi(L) = 0, \quad (104)$$

and therefore has only the zero solution. Hence

$$\ker H_- = \text{span}\{\psi_0\}, \quad \ker H_+ = \{0\}, \quad \text{ind}(Q_\nu) = 1. \quad (105)$$

Since the positive spectra are paired, the entire unpaired compact contribution is the single protected index residue. In determinant language,

$$\log \Delta = \log \Delta_{\text{index}} + \log \Delta_{\text{anomaly}} + \log \Delta_{\text{boundary}} + \log \Delta_{\text{matching}}, \quad (106)$$

with no allowed positive-mode bulk determinant residue.

The index is stable under continuous deformations of f , M_{orb} , or the smooth thin-barrier regulator as long as:

1. Q_ν remains Fredholm;
2. the Q-compatible boundary class is unchanged;
3. no leakage zero mode crosses into the protected sector;
4. anomaly/Ward closure uses the same domain.

This is the protected index residue theorem needed by the framework. The remaining open part is not the kernel algebra; it is proving that the final compact-cell action selects the same Q-compatible domain and anomaly-compatible representation complex.

The determinant and regulator must then be defined from the same compact spectrum:

$$\zeta_i^\pm(s) = \sum_{\lambda_n^\pm > 0} (\lambda_n^\pm)^{-s}, \quad \log \Delta_i = -\zeta_i^-(0) + \zeta_i^+(0). \quad (107)$$

If the positive spectra are exactly paired, the determinant bulk cancels and only index, anomaly, boundary, or frozen matching residues may remain. If the compact domain spoils this pairing, the v0.1 geometry fails the radiative stability route.

35 C3.1e Compact-Resolvent Check For The Relative-Orbifold Domain

The remaining assumption in C3.1d can be checked for the explicit C3.1c relative-orbifold domain. For smooth bounded f on $[0, L]$, the partner Hamiltonians are the regular Sturm–Liouville operators

$$H_- = Q_\nu^\dagger Q_\nu = -\partial_x^2 + \nu^2 f(x)^2 - \nu f'(x), \quad (108)$$

$$H_+ = Q_\nu Q_\nu^\dagger = -\partial_x^2 + \nu^2 f(x)^2 + \nu f'(x). \quad (109)$$

Their domains are the second-order domains induced by $Q_\nu : H^1 \rightarrow L^2$ and $Q_\nu^\dagger : H_0^1 \rightarrow L^2$. The plus sector carries Dirichlet orbifold conditions. The minus sector carries the compatible Robin conditions obtained by requiring $Q_\nu \psi \in H_0^1$:

$$(\partial_x + \nu f)\psi|_{0,L} = 0. \quad (110)$$

These are separated regular boundary conditions on a compact interval. Standard Sturm–Liouville theory therefore gives self-adjoint realizations with compact resolvent:

$$(H_\pm + 1)^{-1} : L^2([0, L]) \rightarrow L^2([0, L]) \quad \text{compact}. \quad (111)$$

Equivalently, the resolvent maps into H^2 with regular boundary conditions, and the embedding $H^2([0, L]) \hookrightarrow L^2([0, L])$ is compact.

Thus C3.1c supplies an explicit domain for which the compact-resolvent assumption in C3.1d is not an extra fitting freedom. What remains open is the parent-selection theorem: the final Tau Core action must still select this relative-orbifold domain and its finite algebra without boundary fitting.

36 C3.1f Variational Boundary-Selection Lemma

The parent-selection problem can be attacked at the boundary level. Consider the positive quadratic wall-domain action

$$S_{\text{dom}}[\psi, \phi] = \|Q_\nu \psi\|^2 + \|Q_\nu^\dagger \phi\|^2 + M_b (|\phi(0)|^2 + |\phi(L)|^2 + |(Q_\nu \psi)(0)|^2 + |(Q_\nu \psi)(L)|^2), \quad M_b > 0. \quad (112)$$

The bulk terms are the same-domain Hessian terms. The boundary term is an orbifold fixed-point obstruction: it penalizes leakage of the plus-sector field and leakage of $Q_\nu \psi$ at the fixed points. Since every term is non-negative, the protected zero-obstruction sector must satisfy

$$\phi(0) = \phi(L) = 0, \quad (Q_\nu \psi)(0) = (Q_\nu \psi)(L) = 0. \quad (113)$$

Thus

$$\phi \in H_0^1([0, L]), \quad (\partial_x + \nu f)\psi|_{0,L} = 0, \quad (114)$$

which are exactly the Dirichlet/Robin boundary data used by the relative-orbifold pair. In that sector the integration-by-parts boundary form vanishes,

$$[\psi^* \phi]_0^L = 0, \quad (115)$$

and the induced second-order domains preserve $Q_\nu : \mathcal{D}(H_-) \rightarrow \mathcal{D}(H_+)$ and $Q_\nu^\dagger : \mathcal{D}(H_+) \rightarrow \mathcal{D}(H_-)$.

This proves a conditional selection lemma: a positive orbifold boundary obstruction in the parent wall action selects the same relative-orbifold domain used in C3.1c–C3.1e. It is not yet the final microscopic proof, because the final parent theory must still derive the existence, coefficient, and universality of S_{dom} rather than inserting it by hand.

37 C3.1g Thin-Barrier Origin Of The Boundary Obstruction

The previous lemma becomes less ad hoc if the boundary obstruction is obtained as a thin-barrier limit of a bulk-localized parent penalty. Let ρ_ϵ be a non-negative approximate delta function supported near the two orbifold fixed points:

$$\rho_\epsilon(x) = \rho_\epsilon^{(0)}(x) + \rho_\epsilon^{(L)}(x), \quad \rho_\epsilon \rightharpoonup \delta_0 + \delta_L. \quad (116)$$

Consider the localized leakage penalty

$$S_{\text{leak},\epsilon} = M_b \int_0^L \rho_\epsilon(x) (|\phi(x)|^2 + |Q_\nu \psi(x)|^2) dx. \quad (117)$$

For fields with continuous representatives at the endpoints, the distributional limit is

$$\lim_{\epsilon \rightarrow 0} S_{\text{leak},\epsilon} = M_b (|\phi(0)|^2 + |\phi(L)|^2 + |(Q_\nu \psi)(0)|^2 + |(Q_\nu \psi)(L)|^2). \quad (118)$$

Thus the C3.1f boundary action is the sharp-orbifold limit of a positive localized leakage energy. The zero-leakage protected sector is again

$$\phi|_{0,L} = 0, \quad Q_\nu \psi|_{0,L} = 0. \quad (119)$$

This gives a plausible parent origin for S_{dom} : it can arise from a finite-width orbifold defect that becomes sharp in the compact-cell limit. The remaining proof obligation is now sharper. The final parent action must derive the defect profile ρ_ϵ , its positivity, its universal coupling M_b , and the absence of competing endpoint terms. If those are not derived, the boundary-selection lemma remains conditional.

38 C3.1h Minimal Defect-Measure Selection Lemma

The defect profile itself can be constrained by admissibility. Let μ_b be the non-negative boundary-localized measure that appears in the leakage term. Assume:

1. locality on the compact interval;
2. invariance under the orbifold reflection exchanging the two fixed points;
3. support only on the fixed-point set of $S^1/\mathbb{Z}_2$;
4. no endpoint flavor label, no Higgs/top endpoint data, and no sign-changing weight.

The fixed-point set is the two-point set $\{0, L\}$. A positive measure supported on it has the form

$$\mu_b = a \delta_0 + b \delta_L, \quad a, b \geq 0. \quad (120)$$

Orbifold exchange symmetry gives $a = b$. Up to the universal coupling M_b , the unique normalized admissible measure is therefore

$$\mu_b = \delta_0 + \delta_L. \quad (121)$$

Equivalently, every admissible smooth regulator family must converge weakly to that measure:

$$\rho_\epsilon(x) dx \rightharpoonup \delta_0 + \delta_L. \quad (122)$$

This proves the measure-level part of the thin-barrier route: once the parent structure requires a positive local leakage penalty supported only at orbifold fixed points, the endpoint support and equal weights are forced. What remains open is the dynamical origin of the leakage penalty and the value/universality of M_b .

39 C3.1i Coupling-Independence Of The Protected Domain

The coefficient M_b is still a parent-scale quantity, but the protected domain selected by the obstruction does not depend on its numerical value as long as it is positive and universal. For the leakage functional

$$S_b[M_b; u] = M_b \int u d\mu_b, \quad u = |\phi|^2 + |Q_\nu \psi|^2, \quad M_b > 0, \quad (123)$$

the zero-obstruction set is

$$S_b[M_b; u] = 0 \iff u = 0 \quad \mu_b\text{-almost everywhere.} \quad (124)$$

Thus every positive value of M_b selects the same protected boundary sector:

$$\phi|_{0,L} = 0, \quad Q_\nu \psi|_{0,L} = 0. \quad (125)$$

Changing M_b changes the gap of boundary-leakage modes, not the protected kernel/index domain. This removes one possible hidden fitting channel: the Higgs quartic, top slot, or endpoint data cannot tune the protected domain through the magnitude of M_b .

The remaining requirement is universality. If M_b is allowed to depend on field species, family, hypercharge, or endpoint observables, the boundary action would reintroduce hidden fitting. The admissible parent route is therefore:

$$M_b = M_{\text{orb}} > 0 \quad (126)$$

as a single orbifold-defect stiffness shared by the wall, cohomology, and regulator sectors. Its numerical value remains an open parent-scale gate, but it is no longer a domain-selection parameter.

40 C3.1j Action-Origin Lemma For The Leakage Term

The preceding gates can be derived from the v0.2 parent-action scaffold if the wall/domain sector contains the universal positive term

$$S_{\text{bdry}} = \epsilon_\tau M_{\text{orb}} \int_0^L (|\Phi_+(x)|^2 + |Q_\nu \Phi_-(x)|^2) d\mu_b(x), \quad M_{\text{orb}} > 0. \quad (127)$$

Here Φ_+ is the plus/parity leakage field and Φ_- is the partner field acted on by the same wall operator Q_ν that defines the Hessian package. Using the measure result of C3.1h,

$$d\mu_b = d\delta_0 + d\delta_L, \quad (128)$$

this term becomes exactly

$$S_{\text{bdry}} = \epsilon_\tau M_{\text{orb}} (|\Phi_+(0)|^2 + |\Phi_+(L)|^2 + |Q_\nu \Phi_-(0)|^2 + |Q_\nu \Phi_-(L)|^2). \quad (129)$$

Variation of a non-negative quadratic action has a zero-obstruction protected sector only when each square vanishes. Hence the action implies

$$\Phi_+|_{0,L} = 0, \quad Q_\nu \Phi_-|_{0,L} = 0, \quad (130)$$

which is precisely the relative-orbifold domain-selection condition.

This is the action-level derivation currently available: if the parent action contains one universal positive fixed-point leakage term, the Q-compatible Dirichlet/Robin domain follows without Higgs/top endpoint tuning. What is still open is one layer deeper: deriving S_{bdry} itself from the final microscopic compact-cell dynamics, rather than accepting it as the v0.2 wall-domain term.

41 C3.1k Compact-Cell To Boundary Reduction Target

The next microscopic route is to make S_{bdry} the thin-layer limit of a compact-cell action, not an independent term. Let r denote a transverse cell coordinate normal to the orbifold fixed set, and let the cell action contain a positive stiffness term for leakage away from the protected parity sector:

$$S_{\text{cell,leak}}[\epsilon] = \epsilon_\tau \int_0^L \int_0^\epsilon (\kappa_\perp^2 |\partial_r \Xi_+(x, r)|^2 + m_\perp^2 |\Xi_+(x, r)|^2 + m_\perp^2 |Q_\nu \Xi_-(x, r)|^2) w_\epsilon(r) dr dx. \quad (131)$$

Assume the transverse profile is gapped and collapses to the fixed set:

$$w_\epsilon(r) dr \rightharpoonup \delta_{\text{fixed}}, \quad \int_0^\epsilon w_\epsilon(r) dr = 1, \quad \kappa_\perp^2, m_\perp^2 > 0. \quad (132)$$

Then the effective fixed-set action has the form

$$S_{\text{cell,leak}} \longrightarrow \epsilon_\tau M_{\text{orb}} \int_0^L (|\Phi_+(x)|^2 + |Q_\nu \Phi_-(x)|^2) d\mu_b(x), \quad (133)$$

where M_{orb} is the transverse gap/stiffness residue. By C3.1h, the admissible fixed-set measure is $d\mu_b = d\delta_0 + d\delta_L$, so the limiting term is precisely S_{bdry} .

This is not yet a solved microscopic compact-cell derivation, because the final theory must still specify the transverse coordinate, the collapse profile w_ϵ , the gapped leakage fields $\Xi_\pm$, and the origin of M_{orb} . But it identifies the exact theorem to prove:

$$S_{\text{cell}} \implies S_{\text{bdry}} \implies \text{Q-compatible relative-orbifold domain.} \quad (134)$$

42 C3.1l Tubular-Neighborhood Collapse Lemma

The transverse coordinate in C3.1k can be made canonical at the level of a compact-cell theorem. Let $F = \{0, L\}$ be the fixed-point set of $S^1/\mathbb{Z}_2$. In a small tubular neighborhood $U_\epsilon(F)$, the distance-to-fixed-set function

$$r(x) = \text{dist}(x, F) \quad (135)$$

is the unique non-negative local coordinate up to smooth reparameterization that is invariant under the orbifold reflection. A localized positive density with no endpoint label can therefore depend only on r :

$$\rho_\epsilon(x) = \frac{1}{Z_\epsilon} \eta\left(\frac{r(x)}{\epsilon}\right), \quad \eta \geq 0, \quad \int \eta = 1. \quad (136)$$

The coarea/tubular-neighborhood reduction gives, for every continuous test function g ,

$$\lim_{\epsilon \rightarrow 0} \int_{U_\epsilon(F)} g(x) \rho_\epsilon(x) dx = g(0) + g(L). \quad (137)$$

Equivalently,

$$\rho_\epsilon(x) dx \rightharpoonup \delta_0 + \delta_L. \quad (138)$$

This derives the collapse profile at measure level from locality, orbifold-invariant distance, positivity, and absence of endpoint labels. The remaining microscopic input is no longer the shape of w_ϵ ; it is the existence of a positive transverse leakage energy whose density is tied to this tubular collapse.

43 C3.1m Stable-Cell Hessian Origin Of Leakage Energy

The remaining microscopic input can be formulated as a stability theorem. Let Ψ_0 be a stationary compact-cell background selected by the parent action $S_{\text{cell}}[\Psi]$, and split small fluctuations into protected and leakage components,

$$\delta\Psi = \delta\Psi_{\text{prot}} \oplus \delta\Psi_{\text{leak}}. \quad (139)$$

If Ψ_0 is a stable minimum modulo gauge/quotient directions, the second variation on the leakage complement is non-negative and gapped:

$$\delta^2 S_{\text{cell}}[\Psi_0](\delta\Psi_{\text{leak}}, \delta\Psi_{\text{leak}}) \geq m_{\perp}^2 \|\delta\Psi_{\text{leak}}\|^2 + \kappa_{\perp}^2 \|\nabla_{\perp} \delta\Psi_{\text{leak}}\|^2, \quad m_{\perp}^2, \kappa_{\perp}^2 > 0. \quad (140)$$

The orbifold quotient identifies leakage away from the protected parity sector with the fields Ξ_+ and $Q_{\nu}\Xi_-$ used in C3.1k. Therefore the quadratic Hessian contains the local positive density

$$\kappa_{\perp}^2 |\partial_r \Xi_+|^2 + m_{\perp}^2 |\Xi_+|^2 + m_{\perp}^2 |Q_{\nu}\Xi_-|^2. \quad (141)$$

Combining this Hessian positivity with the tubular collapse of C3.1l gives

$$S_{\text{cell}} \Rightarrow S_{\text{cell,leak}} \Rightarrow S_{\text{bdry}} \Rightarrow \text{Q-compatible relative-orbifold domain}. \quad (142)$$

This is the strongest current derivation route. It reduces the remaining microscopic gate to a precise stability requirement: prove the existence of a compact-cell stationary background Ψ_0 whose Hessian has a protected sector, a gapped leakage complement, and the same wall operator Q_{ν} on the quotient. If the Hessian has a leakage zero mode, negative mode, or a different operator than Q_{ν} , the compact spectral theorem route fails.

44 C3.1n Explicit Toy Compact-Cell Background And Hessian

An explicit solvable compact-cell model can now be written down. Let the background be a two-component wall field

$$\Psi_0(x) = f_0(x)T_Y, \quad f_0(x) = \tanh x \quad (143)$$

in the local wall chart of the compact cell, with the unoriented identification $T_Y \sim -T_Y$. Take the parent cell energy in the wall sector to contain

$$S_{\text{wall}}[f] = \int \left[\frac{1}{2}(f')^2 + \frac{1}{2}(1-f^2)^2 \right] dx. \quad (144)$$

The Euler–Lagrange equation is

$$-f'' - 2f(1-f^2) = 0, \quad (145)$$

and $f_0(x) = \tanh x$ solves it. The second variation around f_0 gives the wall Hessian

$$L_{\text{wall}} = -\partial_x^2 + 6 \tanh^2 x - 2 = -\partial_x^2 + 4 - 6 \text{sech}^2 x. \quad (146)$$

Its translation zero mode is

$$f'_0(x) = \text{sech}^2 x, \quad (147)$$

which is quotient/collective-coordinate data rather than leakage. Orthogonal to this protected direction, the standard kink Hessian has positive spectrum: one positive bound mode and continuum beginning at 4 in the infinite-line local chart.

The leakage complement is modeled by the positive Hessian block

$$L_{\text{leak}} = -\kappa_{\perp}^2 \partial_r^2 + m_{\perp}^2 + Q_{\nu}^{\dagger} Q_{\nu}, \quad m_{\perp}^2, \kappa_{\perp}^2 > 0. \quad (148)$$

Since $Q_{\nu}^{\dagger} Q_{\nu} \geq 0$, the leakage spectrum obeys

$$\text{Spec}(L_{\text{leak}}) \subseteq [m_{\perp}^2, \infty). \quad (149)$$

Thus this toy compact-cell background has exactly the desired qualitative Hessian structure: a protected collective/quotient sector, a gapped leakage complement, and the same Q_{ν} operator entering the leakage block.

This is still not the final microscopic Tau Core proof. It is an explicit existence model for the Hessian pattern required by C3.1m. The decisive next step is to derive this kink wall sector, leakage block, and Q_{ν} coupling from the final compact-cell action rather than selecting this solvable model.

45 C3.2 What The v0.1 Geometry Would Force

The v0.1 package would upgrade the paper only if the following implications are proven from one domain:

1. the determinant-one unitary sector of the finite algebra gives $S(U(3) \times U(2))$;
2. the traceless block-scalar centralizer is the unique Abelian line and is identified with T_Y after canonical normalization;
3. the orbifold/sign quotient promotes T_Y to the unoriented projector line P_Y ;
4. the lowest finite-tension wall in that line is the Bogomolny kink $f(x) = \tanh x$ in the local-wall limit;
5. the induced wall operator is $Q_i = \partial_x + \kappa_{\tau}^2 Y_i f(x)$;
6. the zero-mode index gives the localization exponent $\nu_i = \kappa_{\tau}^2 |Y_i|$;
7. the same compact domain supplies the zeta/heat-kernel regulator and positive-spectrum pairing needed by the determinant gate.

Items 1–3 are close to finite-dimensional algebra once the algebra is declared. Items 4–7 are the genuine theorem-level work. This is why the v0.1 geometry is important but not yet a completed Higgs derivation.

46 Adversarial Audit Of C3

C3 exposes rather than hides the remaining inputs. The attack points are:

- $S^1/\mathbb{Z}_2$ is still selected as the compact base;
- $\mathbb{C}^3 \oplus \mathbb{C}^2$ is still selected as the finite fiber;
- the even block algebra is still selected as the observable algebra;

- D_{gap} is still declared positive on leakage modes;
- the minimal wall action is still selected;
- the cutoff profile χ is still a regulator input.

The next C4 gate must replace these selections by a parent selection functional. A useful target is

$$\mathcal{S}_{\text{sel}} = I_{\text{compact}} + I_{\text{gap}} + I_{\text{cohom}} + I_{\text{anom}} + I_{\text{wall}} + I_{\text{reg}}, \quad (150)$$

whose minimum or protected extremum should select the C3 data without Higgs/top endpoint information. Until that is shown, C3 is an explicit geometry candidate, not a final derivation.

47 C4 Penalty Interpretation

The selection functional is intended as a sum of non-negative gates. C3 would be selected only if it drives all of them to zero without tuned weights:

- $I_{\text{compact}} = 0$: compact trace-regular spectral data;
- $I_{\text{gap}} = 0$: no extra protected visible sectors;
- $I_{\text{cohom}} = 0$: one primitive closure role and one primitive pairing role;
- $I_{\text{anom}} = 0$: anomaly-safe bridge closure;
- $I_{\text{wall}} = 0$: finite-tension two-vacuum Bogomolny sector;
- $I_{\text{reg}} = 0$: same-domain Q-paired determinant regulator.

C4 is not yet minimized in this ledger. It is a precise target: prove that C3 is a protected extremum, or find a simpler compact geometry that beats it.

48 C4.1 Parent-Selection Lower-Bound Lemma Candidate

C4 can be sharpened into a lower-bound problem. Let admissible compact wall-cell candidates X be scored by

$$\mathcal{S}_{\text{sel}}[X] = I_{\text{compact}}[X] + I_{\text{gap}}[X] + I_{\text{cohom}}[X] + I_{\text{anom}}[X] + I_{\text{wall}}[X] + I_{\text{reg}}[X], \quad I_a[X] \geq 0. \quad (151)$$

The target theorem is not that C3 is attractive, but that every admissible competitor obeys the alternative

$$X \not\sim X_{\text{C3}} \implies \mathcal{S}_{\text{sel}}[X] > 0 \quad (152)$$

unless X is a unitary relabeling, orientation reversal, or Q-exact representative of the same compact domain.

This splits the proof into falsifiable sublemmas:

1. If $I_{\text{compact}} = I_{\text{reg}} = 0$, the wall operator must be a closed Fredholm Q on a compact domain with Q-paired positive spectrum.

2. If $I_{\text{wall}} = 0$, the lowest finite-tension two-vacuum wall is in the Bogomolny class, giving a kink profile in the local-wall limit.
3. If $I_{\text{cohom}} = I_{\text{gap}} = 0$, only two protected visible roles are allowed: one primitive closure role and one primitive pairing role.
4. If $I_{\text{anom}} = 0$, the bridge between those roles must be anomaly-safe under the determinant-one visible unitary sector.

Together these would force the same structure that C3 currently assumes:

$$S^1/\mathbb{Z}_2, \quad \mathbb{C}^3 \oplus \mathbb{C}^2, \quad S(U(3) \times U(2)), \quad Q_i = \partial_x + \kappa_\tau^2 Y_i f(x). \quad (153)$$

This is still a theorem candidate, but it changes the open gate from “choose C3” to “prove a lower bound excluding all lower-complexity competitors.” A single explicit competitor with $\mathcal{S}_{\text{sel}} = 0$ and different protected content would falsify the C4 route.

49 C4.2 No-Free-Weights Selection Rule

The C4 functional must not become a hidden fitting device. Therefore the selection rule is lexicographic rather than tunably weighted:

$$X_1 \prec X_2 \quad \text{iff} \quad (I_{\text{compact}}, I_{\text{reg}}, I_{\text{cohom}}, I_{\text{gap}}, I_{\text{anom}}, I_{\text{wall}})[X_1] < (I_{\text{compact}}, I_{\text{reg}}, I_{\text{cohom}}, I_{\text{gap}}, I_{\text{anom}}, I_{\text{wall}})[X_2] \quad (154)$$

in the first gate where the two tuples differ, with each gate normalized only by its own binary or integer obstruction count. No continuous coefficient may be chosen after seeing Higgs, top, or quartic data.

This turns the parent-selection problem into a strict audit:

- first require compactness and same-domain regulator closure;
- then require the primitive cohomology/gap split;
- then require anomaly-compatible bridge structure;
- only after these discrete gates pass may wall-energy minimality be used to choose among survivors.

The relative-orbifold C3 package is therefore not selected by numerical tuning. It remains viable only if it is the first survivor of this frozen obstruction ordering. This rule is also a falsifier: any lower-complexity competitor with a lexicographically smaller obstruction tuple defeats the present C3 route.

50 C5 Local Spectrum Gate

The local-wall spectrum is now explicit. For

$$Q_\nu = \partial_x + \nu \tanh x, \quad Q_\nu^\dagger = -\partial_x + \nu \tanh x, \quad (155)$$

the partner Hamiltonians are

$$H_- = Q_\nu^\dagger Q_\nu = -\partial_x^2 + \nu^2 - \nu(\nu + 1)\text{sech}^2 x, \quad (156)$$

$$H_+ = Q_\nu Q_\nu^\dagger = -\partial_x^2 + \nu^2 - \nu(\nu - 1)\text{sech}^2 x. \quad (157)$$

The zero mode obeys

$$Q_\nu h_\nu = 0, \quad h_\nu = N_\nu \text{sech}^\nu x. \quad (158)$$

The positive spectra of H_- and H_+ are paired by Q_ν and $Q_\nu^\dagger$, while the unpaired part is the zero-mode/index residue. The local continuum threshold is

$$\lambda_{\text{cont}} = \nu^2. \quad (159)$$

For $\nu_D = 3/10$, this gives

$$\lambda_{\text{cont},D} = 9/100 = 0.090000 \quad (160)$$

This is a real spectrum calculation in the local wall limit. The compact $S^1/\mathbb{Z}_2$ eigenvalue problem, boundary-condition quantization, zeta determinant, and regulator matching remain open.

51 C6 Compact Spectrum Boundary-Value Problem

The compact problem is now formulated. On $x \in [0, L]$, solve

$$H_\pm \psi_n = \lambda_n \psi_n \quad (161)$$

with self-adjoint orbifold boundary conditions that preserve the Q-pairing:

$$Q_\nu : \mathcal{D}(H_-) \rightarrow \mathcal{D}(H_+), \quad Q_\nu^\dagger : \mathcal{D}(H_+) \rightarrow \mathcal{D}(H_-). \quad (162)$$

The compact zeta functions are

$$\zeta_\pm(s) = \sum_{\lambda_n^\pm > 0} (\lambda_n^\pm)^{-s}, \quad (163)$$

and the determinant gate is

$$\log \Delta_\nu = -\zeta'_-(0) + \zeta'_+(0), \quad (164)$$

with zero-mode and index terms kept separately. This is a formulated boundary-value problem, not yet a solved compact determinant. The decisive test is whether the positive spectra remain paired after compact boundary-condition quantization.

52 C7 Competitor Audit

The C7 ledger asks whether the C3 finite fiber is an arbitrary input or the first low-complexity survivor of the C4 penalties. The screen is:

- one-block fibers fail the required closure/pairing split;
- $C^1 \oplus C^1$ is too small for a non-Abelian closure carrier;
- $C^2 \oplus C^2$ has a pairing-like block but no primitive rank-three closure role;
- $C^3 \oplus C^1$ has closure support but no primitive pairing block;
- $C^3 \oplus C^2$ is the first split that carries one closure role and one pairing role with minimal dimensions;
- larger splits are admissible only if an additional gap principle removes the extra protected visible freedom.

Therefore C7 does not prove uniqueness. It converts the hidden $3+2$ choice into a falsifiable minimality statement: C3 must be the first survivor of the C4 cohomology/gap screen, and any lower-complexity survivor would falsify this selection route.

53 C8 Minimality Theorem Candidate

C8 records the theorem-candidate behind the competitor audit. Assume:

1. a finite visible fiber splits into exactly two protected roles, $\mathcal{F} = \mathcal{F}_{\text{cl}} \oplus \mathcal{F}_{\text{pair}}$;
2. the closure role needs a primitive rank-three carrier;
3. the pairing role needs a primitive rank-two carrier;
4. the gap gate forbids extra protected light blocks;
5. the cohomology gate forbids identifying the two roles in a single block.

Then the first admissible finite survivor is

$$\mathcal{F}_{\text{min}} \simeq C^3 \oplus C^2. \quad (165)$$

This is not a final proof because the rank-three closure requirement and the rank-two pairing requirement must still be derived from the microscopic parent action. But it converts the C3 input into a precise target: prove those two minimal role constraints, or find a lower-rank counterexample.

54 C9 Compact Spectrum Pilot

The full compact spectrum remains open, but the packet now contains a first finite-box pilot:

$$H_- = -\partial_x^2 + \nu^2 - \nu(\nu + 1) \text{sech}^2 x, \quad H_+ = -\partial_x^2 + \nu^2 - \nu(\nu - 1) \text{sech}^2 x. \quad (166)$$

The generated CSV `paper4_compact_spectrum_pilot_v01.csv` records the first finite-difference eigenvalues for $H_{\pm}$ at $\nu_D = 3/10$ on a Dirichlet finite box. This is intentionally not promoted to a theorem. It is a reproducible operational pilot whose failure mode is explicit: Dirichlet box boundaries need not preserve the exact $Q_{\nu}, Q_{\nu}^{\dagger}$ pairing. The theorem-level replacement must solve the compact $S^1/\mathbb{Z}_2$ boundary problem with Q-paired self-adjoint domains and then define the zeta determinant from that same spectrum.

55 C9.1 Q-Paired Spectrum Toy Demonstrator

The packet also contains a sharper same-domain toy check: `paper4_q_paired_spectrum_demo_v01.csv`. Here the finite operator Q is built first, and the partner Hamiltonians are defined by

$$H_- = Q^{\dagger}Q, \quad H_+ = QQ^{\dagger}. \quad (167)$$

This changes the logic. The nonzero spectra of $Q^{\dagger}Q$ and $QQ^{\dagger}$ are identical by singular-value decomposition:

$$Qv_n = s_n u_n \implies Q^{\dagger}Qv_n = s_n^2 v_n, \quad QQ^{\dagger}u_n = s_n^2 u_n. \quad (168)$$

The generated CSV records an index residue and a maximum positive-mode pairing residual at numerical precision. This is still not the physical compact orbifold theorem, because the real domain must be the self-adjoint $S^1/\mathbb{Z}_2$ wall domain selected by the parent action. But it shows the right proof mechanism: define Q and its domain first, then the dangerous positive-spectrum determinant cancels structurally, leaving only kernel, index, anomaly, boundary, or frozen matching residues.

56 C10 Uniqueness Proof-Obligation Split

C10 separates the uniqueness problem into layers:

1. finite-dimensional algebra lemmas: rank-three closure minimality, rank-two pairing minimality, and obstruction to merging both roles into one protected block;
2. parent-action selection: no extra protected visible block and no lower-complexity survivor;
3. compact spectral geometry: self-adjoint Q-paired orbifold spectrum;
4. index/anomaly/QFT closure: index residue, Ward identities, regulator consistency, and determinant finiteness on the same domain.

This is why C8 is still a theorem-candidate. The first layer may be attacked as finite-dimensional algebra. The later layers require the explicit parent action and compact spectral geometry. A real uniqueness theorem must not mix proofs from incompatible domains.

57 C11 Algebraic Minimality Lemmas

The first algebraic layer is now explicit:

- if the closure datum is a primitive alternating three-form $\omega_{\text{cl}} \in \Lambda^3 V_{\text{cl}}^*$, then rank one and rank two cannot carry it because $\Lambda^3 V^* = 0$; the minimal closure carrier is C^3 ;
- if the pairing datum is a nondegenerate alternating two-form $\omega_{\text{pair}} \in \Lambda^2 V_{\text{pair}}^*$, then rank one cannot carry it because $\Lambda^2 V^* = 0$; the minimal pairing carrier is C^2 .

Thus, conditional on the parent theory selecting exactly these two primitive protected forms and forbidding extra light blocks, the first finite carrier is $C^3 \oplus C^2$. The remaining open work is not this elementary exterior algebra; it is the parent-action derivation of why these are the protected forms in the first place.

58 C12 Protected-Form Selection Gate

C12 names the next blocker. C11 is conditional on two protected exterior forms:

$$\omega_{\text{cl}} \in \Lambda^3 V_{\text{cl}}^*, \quad \omega_{\text{pair}} \in \Lambda^2 V_{\text{pair}}^*. \quad (169)$$

The parent theory must select these forms through symmetry, cohomology, stationary action, or index obstruction. It is not enough to choose them because they generate the desired $3 + 2$ result. The uniqueness theorem becomes serious only if the same parent structure that selects the forms also supplies the gap principle, compact spectrum, and determinant regulator.

59 C13 Cohomology/Index Selection Route

C13 proposes the first selection route: the two protected forms should be first residues of parent complexes,

$$[\omega_{\text{cl}}] \in H_{\text{cl}}^3, \quad [\omega_{\text{pair}}] \in H_{\text{pair}}^2. \quad (170)$$

If lower-degree residues are trivial or gap-lifted, the degrees three and two are selected by the parent cohomology/index structure rather than chosen after the fact. This remains open until the complexes, differentials, inner product, compact domain, and index theorem are explicitly constructed.

60 C14 Toy Complex Sanity Check

A minimal toy complex can realize the desired pattern:

$$H_{\text{cl}}^k = 0 \ (k < 3), \quad H_{\text{cl}}^3 \simeq C[\omega_{\text{cl}}], \quad H_{\text{pair}}^k = 0 \ (k < 2), \quad H_{\text{pair}}^2 \simeq C[\omega_{\text{pair}}]. \quad (171)$$

This only proves consistency of the target pattern. It does not derive the parent differential, the adjoint, the compact domain, or the index. The next real theorem must replace this toy complex with a parent-selected complex from the compact tau-cell action.

61 C14.1 Anomaly-Bridge Closure Gate

The anomaly gate can also be made sharper without claiming a Standard Model derivation. Let $\mathcal{R}_{\text{vis}}$ be the protected light representation content induced by the compact cell and let T_A denote generators of the visible determinant-one unitary sector. The admissible bridge must satisfy the domain-native trace cancellations

$$\mathcal{A}_{ABC} = \text{Tr}_{\mathcal{R}_{\text{vis}}}(T_A(T_B T_C + T_C T_B)) = 0, \quad \mathcal{A}_A^{\text{grav}} = \text{Tr}_{\mathcal{R}_{\text{vis}}}(T_A) = 0, \quad (172)$$

together with the mixed block condition

$$\mathcal{A}_{3|2} = \text{Tr}_{\mathcal{R}_{\text{bridge}}}(T_3^2 T_2 + T_2^2 T_3) = 0. \quad (173)$$

The point is not the detailed Standard Model spectrum. The point is that the same parent-selected bridge that couples the closure and pairing roles must also be anomaly-neutral in the compact-domain trace. If the 3×2 bridge is removed, the two protected roles become cohomologically disconnected. If an extra unpaired light doublet or singlet is added, at least one trace obstruction must be canceled by an additional parent-derived partner or the candidate fails.

Thus the bridge is not an optional phenomenological decoration. It is a combined cohomology/anomaly gate:

$$I_{\text{anom}} = 0 \implies \text{bridge-closed, trace-neutral, Q-compatible visible content.} \quad (174)$$

This still does not prove anomaly freedom. It states the exact object that the final parent action must compute: a compact-domain representation complex whose protected trace anomalies vanish without adding endpoint-tuned spectator fields.

62 C14.2 Bridge-Trace Normalization Gate

The same bridge gate also constrains the normalization used by the parent Yukawa product. If the protected bridge is exactly one bifundamental channel between the closure block and the pairing block, its multiplicity is not fitted:

$$\dim(V_{\text{cl}} \otimes V_{\text{pair}}^*) = 3 \times 2 = 6. \quad (175)$$

With the compact-cell normalization already fixing $\kappa_\tau^2 = 3/5$, the bridge trace gives

$$g_t^{\text{bridge}} = \text{Tr}_{V_{\text{cl}} \otimes V_{\text{pair}}^*} (\kappa_\tau^2 I) = 6\kappa_\tau^2 = \frac{18}{5}. \quad (176)$$

This is the narrow sense in which the g_t factor is constrained. It is not a physical top Yukawa derivation unless the parent action also derives the unique bridge channel, the family selection, the running/matching factor, and the physical electroweak scale. The falsifier is direct: if the anomaly-safe parent complex requires a different bridge multiplicity or additional light spectator channels, the fixed 18/5 bridge normalization no longer follows. The packet file `paper4_anomaly_bridge_audit_v01.csv` records this as a frozen obstruction table: target bridge, removed bridge, extra unpaired light doublet, and altered bridge multiplicity.

63 C15 Operator-Domain Ansatz

The toy complex must become a closed operator problem. For each branch $a = \text{cl}$ or $a = \text{pair}$, the target data are

$$\mathcal{H}_a = \bigoplus_k \mathcal{H}_a^k, \quad d_a : \mathcal{H}_a^k \rightarrow \mathcal{H}_a^{k+1}, \quad d_a^2 = 0, \quad (177)$$

with adjoint $d_a^\dagger$ and Laplacian

$$\Delta_a = d_a^\dagger d_a + d_a d_a^\dagger. \quad (178)$$

The protected residue must be harmonic and non-exact on the same compact domain used by the wall/regulator/determinant package. This is an operator target, not a solved theorem.

64 C16 Shared-Domain Compatibility Gate

The wall operator, cohomology complex, index, zeta regulator, and determinant must share one compact self-adjoint domain:

$$Q_\nu, d_a, d_a^\dagger, \Delta_a, D_\tau, \zeta_D(s), \det_\zeta. \quad (179)$$

This prevents a false closure in which the zero mode, index, and determinant are each computed with a different boundary condition. C16 is a compatibility gate; the actual domain classification remains open.

65 C16.1 Anomaly-Compatible Domain Closure Criterion

The anomaly/Ward gate can be made domain-native. Let $\mathcal{D}_{\text{rel}}$ be the Q-compatible relative-orbifold domain selected above. A gauge, BRST, or cohomology generator G_A is admissible only if

$$G_A \mathcal{D}_{\text{rel}} \subseteq \mathcal{D}_{\text{rel}}, \quad [G_A, Q_\nu] \mathcal{D}_{\text{rel}} \subseteq L^2([0, L]), \quad (180)$$

and if its regulated trace is computed with the same compact spectrum:

$$\mathcal{A}_{ABC} = \lim_{\Lambda \rightarrow \infty} \text{Tr}_{\mathcal{D}_{\text{rel}}} [T_A (T_B T_C + T_C T_B) \chi(D_\tau^2/\Lambda^2)]. \quad (181)$$

Thus anomaly cancellation is not an abstract representation statement detached from the wall problem. It must be a trace statement on the same domain that defines Q_ν , the index, and the determinant.

This gives the closure criterion:

$$I_{\text{anom}} = 0 \iff \begin{cases} G_A \mathcal{D}_{\text{rel}} \subseteq \mathcal{D}_{\text{rel}}, \\ \mathcal{A}_{ABC} = 0, \\ \mathcal{A}_A^{\text{grav}} = 0, \\ \text{no added endpoint spectator fields.} \end{cases} \quad (182)$$

If a would-be anomaly cancellation requires changing the boundary condition, adding a branch-specific regulator, or adding endpoint spectator modes, it is not compatible with the compact spectral theorem. This is still a criterion, not a completed anomaly calculation; the parent action must still derive the representation complex whose generators satisfy it.

66 C16.3 UV/Continuum Admissibility Criterion

The compact spectral mechanism can be promoted toward QFT only if it admits a controlled continuum family. Let $\epsilon > 0$ denote a compact-cell cutoff, defect thickness, lattice spacing, or spectral truncation scale, and let

$$(\mathcal{H}_\epsilon, Q_\epsilon, D_\epsilon, \chi_\epsilon, \mathcal{D}_\epsilon)$$

be the corresponding regulated data. The continuum gate requires:

1. Q_ϵ converges to a closed Fredholm Q in graph norm or strong-resolvent sense;
2. the domains $\mathcal{D}_\epsilon$ converge to the same relative-orbifold domain used by the index and determinant gates;
3. the index is stable, $\text{ind}(Q_\epsilon) = \text{ind}(Q)$, for sufficiently small ϵ ;
4. the positive-spectrum determinant residue is regulator independent after C18–C20 pairing;
5. the low-energy projected action has local kinetic terms, finite renormalized couplings, and the same Ward/anomaly constraints as C16.1–C16.2.

This is a UV/continuum admissibility criterion, not a microscopic completion. It changes the open problem from “invent a UV theory” to a sharper test: construct a regulator family whose limit preserves the compact index, same-domain determinant cancellation, and anomaly/Ward identities.

67 C16.3a Microscopic Continuum Completion Target

The continuum blocker can be narrowed further by specifying the kind of microscopic family that would close it. A candidate completion is a sequence of compact-cell actions

$$S_\epsilon[\Psi] = \int_{K_\tau^\epsilon} (\langle D_\epsilon \Psi, D_\epsilon \Psi \rangle + V_\epsilon(\Psi) + B_\epsilon(\Psi | \partial K_\tau^\epsilon)) d\mu_\epsilon, \quad (183)$$

with stationary backgrounds Ψ_0^ϵ satisfying

$$\delta S_\epsilon[\Psi_0^\epsilon] = 0, \quad \delta^2 S_\epsilon[\Psi_0^\epsilon] \longrightarrow H_\tau \quad (184)$$

in strong-resolvent or graph-norm sense. The completion gate is passed only if the Hessian limit produces the same Q_ν domain, the same index residue, the same trace-cancellation complex, and the same determinant residue class:

$$(S_\epsilon, \Psi_0^\epsilon) \longrightarrow (\mathcal{D}_{\text{rel}}, Q_\nu, \text{ind } Q_\nu, \mathcal{R}_{\text{vis}}, R_t^{\text{fin}}). \quad (185)$$

This forbids a continuum story in which the wall, anomaly cancellation, regulator, and determinant are imported from separate effective models. The paper has not supplied such a microscopic family, but the target is now mathematically explicit.

68 C16.2 Same-Domain Ward Identity Gate

C16.1 constrains the anomaly trace. The next closure condition is a Ward identity on the same regulated compact domain. Let J_A be the current generated by an admissible G_A , and let S_{eff}^Λ denote the finite-cutoff effective action obtained from the same operator family D_τ^2 used in the determinant gate. The required identity is

$$\delta_A S_{\text{eff}}^\Lambda = \langle \nabla_\mu J_A^\mu \rangle_\Lambda = \mathcal{A}_A^\Lambda + \mathcal{B}_A^\Lambda, \quad (186)$$

where $\mathcal{A}_A^\Lambda$ is the heat-kernel anomaly term and $\mathcal{B}_A^\Lambda$ is a possible boundary/domain variation. The domain gate demands

$$\lim_{\Lambda \rightarrow \infty} \mathcal{A}_A^\Lambda = 0, \quad \lim_{\Lambda \rightarrow \infty} \mathcal{B}_A^\Lambda = 0, \quad (187)$$

without changing $\mathcal{D}_{\text{rel}}$ and without adding endpoint counterterms that are absent from the parent action. Equivalently, the admissible regulator must commute with the protected positive-mode pairing up to trace-class terms:

$$[\chi(D_\tau^2/\Lambda^2), Q_\nu] = \mathcal{O}_{\text{tr}}(\Lambda^{-1}), \quad \lim_{\Lambda \rightarrow \infty} \text{Tr}_{\mathcal{D}_{\text{rel}}}([G_A, Q_\nu]\chi(D_\tau^2/\Lambda^2)) = 0. \quad (188)$$

If this fails, the determinant cancellation may be a formal pairing artifact rather than a regulator-safe QFT statement. This gives a conditional same-domain anomaly/Ward closure criterion: the trace and Ward terms must vanish on the same compact domain used by Q_ν , the index, and the determinant.

69 C16.4 Representation Trace-Cancellation Gate

The remaining anomaly problem can now be stated as a finite trace-cancellation problem rather than as an informal representation hope. A parent-derived visible complex must provide a finite set of compact-domain modules

$$\mathcal{R}_{\text{vis}} = \{(\mathcal{H}_a, \chi_a, T_A^a, \mathcal{D}_a, \sigma_a)\}_a, \quad (189)$$

where χ_a is chirality or orientation, $\sigma_a = \pm 1$ is the determinant sign, and every $\mathcal{D}_a$ is induced from the same compact-cell domain. The admissible trace-cancellation condition is

$$\mathcal{T}_{ABC}^{\text{parent}} = \sum_a \sigma_a \text{Tr}_a[\chi_a T_A^a (T_B^a T_C^a + T_C^a T_B^a)] = 0, \quad (190)$$

together with the gravitational and mixed trace conditions

$$\sum_a \sigma_a \text{Tr}_a(\chi_a T_A^a) = 0, \quad \sum_a \sigma_a \text{Tr}_a(\chi_a) = 0. \quad (191)$$

The crucial restriction is that $\mathcal{R}_{\text{vis}}$ may not be chosen after the anomaly is known. It must be the image of the same parent selection that produces the $3+2$ carrier, the hypercharge line, and the wall domain:

$$S_{\text{parent}} \longrightarrow (\mathcal{F}, L_Y, \mathcal{D}_{\text{rel}}, \mathcal{R}_{\text{vis}}). \quad (192)$$

If extra spectator modules are inserted solely to cancel $\mathcal{T}_{ABC}^{\text{parent}}$, the closure fails. This section therefore turns the old open phrase “derive the representation trace cancellation” into a concrete gate: compute $\mathcal{R}_{\text{vis}}$ from the parent compact cell and verify the three traces above on the same domain. The trace equations are now fixed, but the parent-derived representation complex is not yet computed.

70 C17 Self-Adjoint Domain Candidate

The first candidate replacement for the C9 Dirichlet pilot is an orbifold parity domain that preserves Q-pairing:

$$Q_\nu : \mathcal{D}_- \rightarrow \mathcal{D}_+, \quad Q_\nu^\dagger : \mathcal{D}_+ \rightarrow \mathcal{D}_-. \quad (193)$$

In the relative-orbifold candidate the domains are made explicit as

$$\mathcal{D}_- = \{\psi \in H^1([0, L]) : \psi(0) = \psi(L) = 0\}, \quad (194)$$

$$\mathcal{D}_+ = \left\{ \phi \in H^1([0, L]) : (Q_\nu^\dagger \phi)(0) = (Q_\nu^\dagger \phi)(L) = 0 \right\}. \quad (195)$$

The second-order domains are then not chosen independently:

$$\mathcal{D}(H_-) = \{\psi \in \mathcal{D}_- : Q_\nu \psi \in \mathcal{D}_+\}, \quad (196)$$

$$\mathcal{D}(H_+) = \left\{ \phi \in \mathcal{D}_+ : Q_\nu^\dagger \phi \in \mathcal{D}_- \right\}. \quad (197)$$

This is the important restriction. The boundary condition of $H_\pm$ is inherited from the first-order closed operator pair rather than fitted at the second-order level. It must make $H_\pm$ self-adjoint, preserve positive-mode pairing, keep zero-mode/index residue explicit, and feed the same spectrum to the zeta determinant. This is a domain candidate, not a classification theorem.

71 C17.1 Parent-Domain Selection Functional

The remaining parent-domain question can be sharpened into a selection functional on admissible self-adjoint extensions. Let $\mathfrak{D}_Q$ be the class of compact domains for which Q_ν is closed and $H_\pm$ are self-adjoint. Define the lexicographic obstruction vector

$$\mathcal{S}_{\text{dom}}(\mathcal{D}) = (I_Q(\mathcal{D}), I_{\text{ind}}(\mathcal{D}), I_{\text{pair}}(\mathcal{D}), I_{\text{bdry}}(\mathcal{D}), I_{\text{anom}}(\mathcal{D}), I_{\text{free}}(\mathcal{D})), \quad (198)$$

where the entries penalize, respectively: failure of Q -closure, wrong index, unpaired positive spectrum, unmatched boundary residue, anomaly/Ward-domain failure, and any continuous endpoint parameter not fixed by the parent action. The parent-domain selection rule is

$$\mathcal{D}_{\text{parent}} = \arg \min_{\mathcal{D} \in \mathfrak{D}_Q} \mathcal{S}_{\text{dom}}(\mathcal{D}) \quad \text{lexicographically.} \quad (199)$$

The relative-orbifold domain is selected if it is the unique zero-obstruction minimum:

$$\mathcal{S}_{\text{dom}}(\mathcal{D}_{\text{rel}}) = (0, 0, 0, 0, 0, 0) \quad (200)$$

and every competitor either changes the index, breaks Q -pairing, introduces an endpoint parameter, or fails the anomaly/Ward domain criterion. This closes the domain-selection problem at criterion level, not at exhaustive classification level: a lower-complexity zero-obstruction competitor would still falsify the claimed parent selection.

72 C18 Paired-Spectrum Verification Target

The required theorem is the positive-mode bijection

$$H_- \psi_n = \lambda_n \psi_n, \quad \lambda_n > 0 \quad \implies \quad \chi_n = \lambda_n^{-0.5} Q_\nu \psi_n, \quad H_+ \chi_n = \lambda_n \chi_n, \quad (201)$$

with the converse map through $Q_\nu^\dagger$. Only kernel terms may remain unpaired:

$$\text{ind}(Q_\nu) = \dim \ker Q_\nu - \dim \ker Q_\nu^\dagger. \quad (202)$$

The proof obligation is now narrow. If Q_ν is closed Fredholm on the C17 domain and $H_- = Q_\nu^\dagger Q_\nu$, $H_+ = Q_\nu Q_\nu^\dagger$ are self-adjoint with compact resolvent, then for every $\lambda > 0$:

$$Q_\nu : \ker(H_- - \lambda) \longrightarrow \ker(H_+ - \lambda) \quad (203)$$

is injective because $Q_\nu \psi = 0$ would imply $\lambda \|\psi\|^2 = \|Q_\nu \psi\|^2 = 0$, and it is surjective by the inverse map $\lambda^{-0.5} Q_\nu^\dagger$. Thus the positive spectra match including multiplicity:

$$\text{Spec}_+(H_-) = \text{Spec}_+(H_+). \quad (204)$$

This proves the abstract pairing lemma conditional on the C17 domain being the actual compact self-adjoint domain selected by the parent action. What remains open is not the algebraic pairing step; it is the domain-selection theorem and the anomaly/Ward-compatible regulator.

73 C19 Conditional Zeta-Determinant Cancellation Lemma

If C18 holds on the same compact domain, then the positive-mode determinant ratio should cancel:

$$\log \frac{\det_\zeta H_-^{\text{pos}}}{\det_\zeta H_+^{\text{pos}}} = -\zeta'_{-, \text{pos}}(0) + \zeta'_{+, \text{pos}}(0) = 0. \quad (205)$$

The determinant lemma is then immediate at the formal zeta level. Define

$$\zeta_\pm^{\text{pos}}(s) = \sum_{\lambda \in \text{Spec}_+(H_\pm)} m_\pm(\lambda) \lambda^{-s}. \quad (206)$$

If C18 gives $m_-(\lambda) = m_+(\lambda)$ for all positive eigenvalues and the same analytic continuation/regulator is used, then

$$\zeta_-^{\text{pos}}(s) = \zeta_+^{\text{pos}}(s) \quad \implies \quad \det_\zeta H_-^{\text{pos}} = \det_\zeta H_+^{\text{pos}}. \quad (207)$$

Any remaining contribution must be explicit kernel/index/anomaly/boundary or matching residue. This establishes the positive-spectrum determinant cancellation proposition inside the same-domain setup. The remaining open determinant problem is physical rather than algebraic: compute the top-sensitive same-domain finite residue, prove Ward/regulator compatibility, and match the result to the low-energy Higgs sector without inserting a counterterm.

74 C20 Residue Classification Gate

After positive-mode cancellation, allowed residues are only those fixed by the same domain-native structure: kernel/index, anomaly/Ward, boundary, or frozen matching terms. Forbidden residues include endpoint-tuned scalar counterterms, branch-specific regulators, unmatched boundary constants, or arbitrary finite terms inserted to restore naturalness. Thus cancellation is not enough; the remaining finite residue must also be admissible.

75 C20.1 Regulator-Independence Residue Test

The residue classification becomes testable by comparing two admissible cutoff functions. Let χ_1 and χ_2 be smooth compact-domain regulators with $\chi_i(0) = 1$ and rapid decay. The positive-mode determinant difference is admissible only if

$$\Delta_{12}^{\text{pos}} = \text{Tr}_{\mathcal{D}_{\text{rel}}} [(\chi_1 - \chi_2)(D_\tau^2/\Lambda^2) (\Pi_-^{\text{pos}} - \Pi_+^{\text{pos}})] \xrightarrow{\Lambda \rightarrow \infty} 0. \quad (208)$$

Here $\Pi_\pm^{\text{pos}}$ are the positive-spectrum projectors of $H_\pm$ on the same domain. If C18 holds exactly, Π_-^{pos} and Π_+^{pos} are matched by Q_ν , and no bulk positive-mode residue may depend on χ_i .

The allowed finite ambiguity is therefore restricted to

$$\Delta_{12} \in \text{span} \{ \text{index, anomaly, boundary, frozen matching} \}. \quad (209)$$

Any regulator-dependent scalar mass term outside this span is a failure mode. This is the practical audit version of C20: change the admissible cutoff, and the Higgs-lightness conclusion must not move except through the explicitly listed residues.

76 ϵ_τ Origin Gate

The last absolute scale is ϵ_τ . Admissible origins include:

- a universal parent stiffness/density scale;
- a quantized compact-cell action unit;
- a parent-derived wall-tension unit;
- an independent gravity/cosmology calibration frozen before Higgs/top comparison.

Forbidden origins include Higgs-vev backsolve, top-mass backsolve, particle-specific ϵ_τ , Branch-A-only scale insertion, or an unmatched Planck-ratio insertion.

77 Wall-Cell Embedding Gate

The Branch A wall domain must be related to the compact cell by

$$\mathcal{W}_A \subseteq \mathcal{K}_\tau \quad \text{or} \quad \mathcal{W}_A = \mathcal{K}_\tau / \mathcal{H}_A. \quad (210)$$

The induced measure must come by restriction, quotient, or coarea. The Higgs and top operators must use the same domain and regulator. The count N_τ must be compatible with wall index/residue bookkeeping. Otherwise the compact-cell unit and Higgs wall remain disconnected hidden scales.

78 v0.2 Single-Package Parent Action

The current strongest scaffold is

$$S_A^{(0.2)} = \epsilon_\tau [S_{\text{cell}} + S_{\text{wall}} + S_{\text{gap}} + S_Q + S_{\text{anom}} + S_{\text{prod}} + S_{\text{reg}}]. \quad (211)$$

It keeps in one package:

- compact tau cell and count sector;
- induced Branch A wall domain;
- trace quotient;
- unique hypercharge wall channel;
- positive leakage gaps;
- minimal P, C, B cohomology roles;
- same-domain product and regulator;
- universal ϵ_τ ;
- fixed dimensionless y_t^{parent} .

This is the current best single scaffold, but still not the final action.

79 Top Determinant Cancellation Candidate

The radiative gate requires the physical top-sensitive operator to live on the same wall-domain Q package. If positive spectra of $Q_t^\dagger Q_t$ and $Q_t Q_t^\dagger$ are paired and the regulator is shared, then the signed positive-spectrum determinant can cancel structurally. Zero modes and index residue must remain explicit. The gate is not solved until the physical Q_t , determinant signs, regulator, and Yukawa strength are parent-derived.

80 Top Determinant No-Mass-Rescue Gate

The decisive radiative-stability test is not whether a formal determinant can be written. It is whether the top-sensitive determinant avoids producing an unprotected scalar mass term. Let H denote the visible Higgs zero-mode coordinate and let δ denote the top/flavor mismatch or bridge spurion. The one-loop effective contribution must have the constrained expansion

$$\Gamma_t[H, \delta] = c_0 + c_2^{\text{prot}} |H|^4 + \delta \mathcal{O}_{\text{bridge}}[H] + \mathcal{O}(\delta^2), \quad (212)$$

with the forbidden terms

$$\Delta\Gamma_t^{\text{forbidden}} = M_\tau^2 |H|^2 + M_\tau \delta |H|^2 + c_{\text{free}} \Lambda^2 |H|^2. \quad (213)$$

Each forbidden term must be either absent, Q -exact, quotient-null, or paired away by the same-domain determinant:

$$\Pi_{\text{phys}} \Delta\Gamma_t^{\text{forbidden}} = 0. \quad (214)$$

This is the no-mass-rescue gate. A successful top determinant may generate a controlled quartic/matching residue or a parent-derived Yukawa coefficient, but it may not hide the hierarchy problem in a mismatch-independent scalar mass. If the calculation requires a counterterm chosen to cancel $M_\tau^2 |H|^2$, the Branch A Higgs-lightness mechanism fails.

81 Top Determinant Mass-Derivative Test

The no-mass-rescue gate can be expressed as a derivative test on the regulated determinant. Let the top-sensitive squared operator be

$$\mathcal{O}_t(H, \delta) = \mathcal{O}_0 + \delta \mathcal{O}_1 + H \mathcal{Y} + H^\dagger \mathcal{Y}^\dagger + |H|^2 \mathcal{M}_2 + \dots, \quad (215)$$

all on the same compact domain and with the same regulator as C16–C20. Define

$$\Gamma_t^\Lambda(H, \delta) = -\frac{1}{2} \text{Tr}_{\mathcal{D}_{\text{rel}}} \log(\mathcal{O}_t(H, \delta)/\Lambda^2)_\Lambda. \quad (216)$$

The forbidden Higgs mass coefficient is the local curvature at the quotient origin:

$$\mu_t^2(\delta) = \left. \frac{\partial^2 \Gamma_t^\Lambda}{\partial H^\dagger \partial H} \right|_{H=0}. \quad (217)$$

The gate requires

$$\mu_t^2(0) = 0, \quad \left. \frac{\partial \mu_t^2}{\partial \delta} \right|_{\delta=0} = 0 \quad (218)$$

unless the resulting term is in the already allowed index/anomaly/boundary/frozen-matching residue class. In trace form the dangerous term is

$$\mu_t^2 \sim \text{Tr} \left(\mathcal{O}_0^{-1} \mathcal{M}_2 - \mathcal{O}_0^{-1} \mathcal{Y} \mathcal{O}_0^{-1} \mathcal{Y}^\dagger \right)_{\text{same domain}}. \quad (219)$$

Thus the next real calculation is concrete: compute this same-domain trace and show that it vanishes, is quotient-trivial, or reduces to a permitted fixed residue. If it is nonzero and regulator-dependent, the Higgs-lightness route fails even if the quartic overlap remains elegant.

The reproducibility packet includes the toy audit `paper4_top_mass_derivative_toy_trace_v01.csv`. It contains two deliberately simple cases. In the paired same-domain case the forbidden mass curvature cancels mode-by-mode. In the unpaired regulator-failure case a single shifted positive mode leaves a nonzero curvature. This is only a toy trace, but it fixes the diagnostic signature required of the physical top determinant.

The same file also includes a mismatch-sensitivity scan. The scan shifts one positive mode by a controlled fraction and records the induced forbidden curvature. This is useful because the real determinant proof must not merely cancel a perfectly symmetric toy example; it must show that any nonzero domain/regulator mismatch is either absent by theorem or appears as a detected failure mode.

Finally, the toy file compares two cutoff families, exponential and rational. In the paired case both regulators keep the forbidden curvature at zero. In the shifted case the curvature becomes cutoff-dependent. This is the intended C20.1 behavior: regulator choice should be invisible only after the same-domain positive spectra have really paired.

82 Top Determinant Finite-Residue Extraction Gate

The previous sections show what must cancel. The remaining physical question is narrower: after same-domain positive-mode cancellation, what finite top-sensitive residue is still admissible? Define the regulated signed determinant difference by

$$\Delta \Gamma_t^\Lambda(H) = \Gamma_{t,-}^\Lambda(H) - \Gamma_{t,+}^\Lambda(H), \quad (220)$$

Top mass-derivative toy stress test

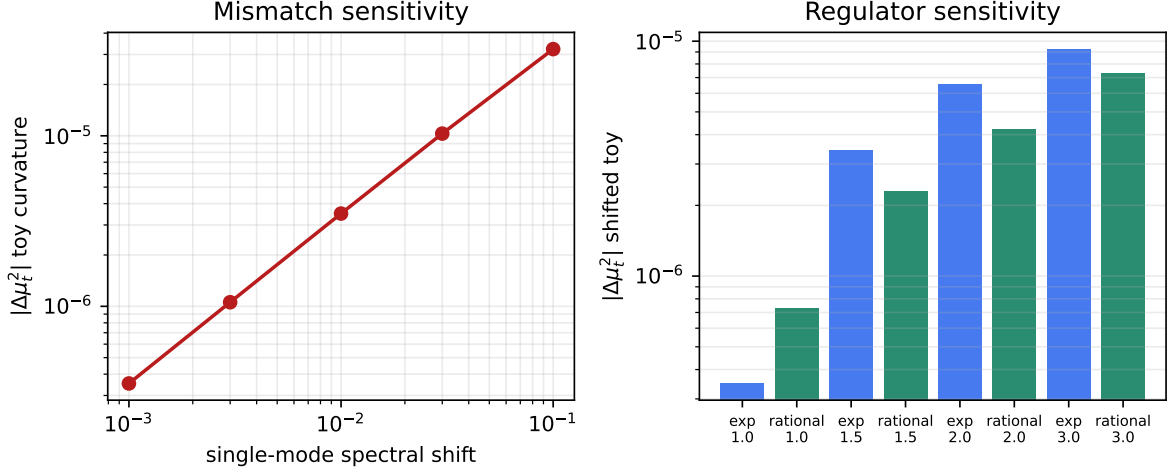


Figure 1: Toy top-mass derivative stress test. Left: a single positive-mode mismatch induces a growing forbidden curvature. Right: paired spectra remain zero under two cutoff families, while shifted spectra leave regulator-dependent curvature. This is a diagnostic toy audit, not the physical top determinant, and is not used as independent evidence for radiative stability.

and subtract the parts already fixed by the paired positive spectrum, kernel/index sector, anomaly/Ward sector, and boundary sector:

$$R_t^{\text{fin}}(H) = \text{FP}_{\Lambda \rightarrow \infty} \left[\Delta\Gamma_t^\Lambda(H) - \Delta\Gamma_t^{\text{pos}}(H) - \Delta\Gamma_t^{\text{ind/anom/bdry}}(H) \right]. \quad (221)$$

Here FP denotes the regulator-independent finite part. The admissible residue condition is

$$R_t^{\text{fin}} \in \text{span} \{ R_{\text{index}}, R_{\text{anomaly}}, R_{\text{boundary}}, R_{\text{frozen matching}} \}. \quad (222)$$

Any remaining contribution outside this span is not a harmless finite correction; it is an unaccounted physical term. In particular, the Higgs-mass curvature

$$\mu_{t,\text{fin}}^2 = \left. \frac{\partial^2 R_t^{\text{fin}}}{\partial H^\dagger \partial H} \right|_{H=0} \quad (223)$$

must vanish, be quotient-null, or be a fixed parent-derived matching residue. It may not be adjusted after the endpoint is known. This is the finite-residue extraction gate: the paper now defines the required computation, but it does not yet compute the physical R_t^{fin} from a complete parent QFT.

83 Toy Demonstrator

The toy demonstrator sets

$$\epsilon_\tau = \mu_\tau(\mathcal{K}_\tau) = N_\tau = C_A = \hat{E}_{\text{wall}} = \hat{M}_{\text{stab}} = g_{\text{vis}} = R_t = 1. \quad (224)$$

Then

$$m_{\text{toy}} = \frac{y_t^{\text{parent}}}{\sqrt{2}} = 0.6799201036561399. \quad (225)$$

This is finite and endpoint-free, but it has no GeV interpretation.

84 Audit Map

The generated repository checks:

- the gamma-function quartic overlap and sensitivity scan;
- Wolfram checks for normalization, hypercharge, line quotient, BRST skeleton, and full-derivation gate-ledger coverage;
- packet CSV values for $I_4(3/10)$, 0.267097842200176, $g_t = 18/5$, and y_t^{parent} ;
- TeX/PDF regeneration and arXiv source package creation;
- explicit claim-boundary strings in the manuscript.

85 Full Gate Ledger

Gate	Current status
$3 + 2$ stabilizer	derived in minimal Branch A carrier
$\kappa_\tau^2 = 3/5$	canonical hypercharge normalization
unoriented line quotient	theorem-candidate with projector audit
$\tanh x$ wall	minimal/BPS wall route; parent derivation open
$\nu_i = \kappa_\tau^2 Y_i $	conditional F1–F8 forcing route
quartic overlap	computed: $I_4(3/10) = 0.133756368895144$
Yukawa overlap	computed: $\mathcal{I}_{qHu} = 0.267097842200176$
g_t	bridge-trace candidate 18/5
y_t^{parent}	fixed dimensionless coefficient 0.961552231920634
v_{eff}	stabilizer-gap route, not measured vev derivation
R_t	running/matching shape gate
ϵ_τ	last universal absolute scale blocker
wall-cell embedding	same-domain compact-cell requirement
top determinant	positive-spectrum cancellation plus finite-residue extraction gate; physical residue not computed
BRST/anomaly/regulator	same-domain Ward and trace-cancellation gates formulated;
	parent complex open
Standard Model emergence	roadmap only

86 What Would Count As A Win

The Higgs route would become much stronger if a single final parent action:

1. derives the two protected visible modules;
2. derives F1–F8 rather than assuming them;
3. selects $\mathcal{K}_\tau$ and $\mathcal{W}_A$ with the required embedding;
4. fixes ϵ_τ and C_A without Higgs/top endpoint data;

5. closes the BRST/anomaly/regulator package;
6. computes the physical top finite residue and matching map.

87 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS} .

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

87.1 Inherited contact and the occupied-state distinction

For a positive symmetric three-dimensional quotient K , define $t_K = \text{tr } K$ and $s_K = \text{tr } K^2$. The BRAC shape invariant and its differential are

$$I_K = 9s_K/t_K^2 - 3, \quad (226)$$

$$dI_K[\delta K] = \text{tr} \left[(18K/t_K^2 - 18s_K \mathbf{1}/t_K^3) \delta K \right]. \quad (227)$$

If $K = (LH^{-1}L^*)^{-1}$ on the nonsingular leaf, then $\delta K = -K\delta(LH^{-1}L^*)K$. Common scaling and orthogonal frame rotation have zero dI_K ; the existing spectral-shape control has $I'_K(0) = -1.06341377025\dots$. This does not identify its coordinate with a physical apparatus displacement.

For one real canonically normalized field at fixed coframe and length,

$$S_K = -\frac{1}{2} \int \sqrt{-g} \frac{I_K}{\ell_H^2} \phi^2 d^4x, \quad b_j = \ell_H^{-2} dI_K[\partial_{a_j} K]. \quad (228)$$

The corresponding potential interaction is $H_{\text{int}} = \sum_j a_j B_j$ with $B_j = \frac{1}{2} \int b_j \phi^2 d^3x$. The actual subsystem pullback, kinetic terms and calibration remain required. For a free real scalar in a stationary inertial vacuum the Wick-square correlation is $2W^2$, so this contact has a two-quantum threshold $2\nu_B$, not the linear comparator's ν_B . With $c = \hbar = \ell_H = 1$, $m_0 = 0$ and the source CHDF value $I_K = 0.0610078751932586$, the thresholds are 0.24699772305278 and 0.49399544610556. The previous comparison gaps 0.35 and 0.15 therefore have different support in the two contact models. Neither threshold is asserted for arbitrary trajectories or occupied states.

For a supplied coherent background, mean force and connected covariance follow from the same contact:

$$\langle B_i \rangle = \frac{1}{2} \int b_i \phi_{\text{cl}}^2 d^3x, \quad (229)$$

$$C_{ij}(t, t') = \int d^3x d^3y b_i(x) b_j(y) \left[\phi_{\text{cl}}(t, x) \phi_{\text{cl}}(t', y) W(t, x; t', y) + \frac{1}{2} W(t, x; t', y)^2 \right]. \quad (230)$$

Vacuum Wick ordering here is a comparison prescription, not a derivation of the physical stress renormalization. A nonzero classical mean does not by itself select this quantum state. The kernel is a free interaction-picture input, generally nonstationary, not an all-orders interacting bath law.

87.2 Conditional state selection and its resource requirement

For a supplied finite positive oscillator packet, fixed means $\text{Tr}(\rho c_r) = \alpha_r$ imply

$$\langle H_{\text{exc}} \rangle = \sum_r \hbar \omega_r |\alpha_r|^2 + \sum_r \hbar \omega_r \langle (c_r - \alpha_r)^\dagger (c_r - \alpha_r) \rangle. \quad (231)$$

The second sum is nonnegative. Equality with the classical mean energy forces the state into the common kernel of all $c_r - \alpha_r$, hence the unique coherent state. With excess energy ΔE ,

$$1 - \langle \alpha | \rho | \alpha \rangle \leq \frac{\Delta E}{\hbar \omega_{\min}}. \quad (232)$$

Displaced number states have the same means and higher energy, providing an explicit counterexample when the zero-excess condition is omitted. Ground-state minimization does not provide autonomous preparation. For energy-calibrated amplitude ψ_E , the source-preserving displacement is $\alpha = (2\hbar\Omega)^{-1/2} \psi_E$. If the input is action-normalized, the energy/clock conversion remains necessary; it cannot be replaced by setting every mode frequency to one.

For a declared common-frame tetrahedral coherent code, equal priors and $\alpha_i = \sqrt{N} n_i$, $n_i \cdot n_j = -1/3$, the off-diagonal overlap is $s = \exp(-4N/3)$. The exact optimal single-shot discrimination probability is

$$P_{\text{opt}} = \frac{[\sqrt{1+3s} + 3\sqrt{1-s}]^2}{16}. \quad (233)$$

The square-root measurement attains this value. With $S = \sqrt{G}$ and $d = S_{ii}$, the dual operator $Y = dS/4$ has matching trace and $Y - v_i v_i^*/4 \succeq 0$ for columns v_i of S , proving optimality. Uniform pure loss replaces N by ηN only in that declared channel model. This coherent code is not the earlier one-particle leaf encoding; the bound does not construct a physical receiver or stable Q_{OS} .

87.3 Joint coupling fixes local uncertainty

The existing two-copy action, expanded at an already aligned local well, has $H = \begin{pmatrix} D & -C \\ -C^T & D \end{pmatrix}$, $D = e_*(S_i + 16\mathbf{1})$, $C = 16e_*\mathbf{1}$ and $S_i = 3P_i + 4(\mathbf{1} - P_i)$. Given a positive kinetic matrix G , canonical quantization and joint harmonic ground preparation, put $W = (G^{-1/2}HG^{-1/2})^{1/2}$. Normal-mode diagonalization gives

$$V_z = \frac{\hbar}{2}G^{-1/2}W^{-1}G^{-1/2}, \quad V_p = \frac{\hbar}{2}G^{1/2}WG^{1/2}. \quad (234)$$

For equal inertia I_0 , a ray of curvature $s_0 = 3, 4$ has frequencies $\omega_c^2 = e_*s_0/I_0$ and $\omega_r^2 = e_*(s_0 + 32)/I_0$. Its local symplectic eigenvalue is

$$\nu_{s_0} = \frac{1}{4}\sqrt{2 + \omega_r/\omega_c + \omega_c/\omega_r} > \frac{1}{2}. \quad (235)$$

The joint state is pure but the equal-inertia three-mode local purity is 0.627817314635553. The local position covariance also equals

$$V_x = \frac{\hbar}{\pi} \int_0^\infty [D + I_0\xi^2\mathbf{1} - C(D + I_0\xi^2\mathbf{1})^{-1}C^T]^{-1} d\xi. \quad (236)$$

Thus the full frequency-dependent Schur kernel, not its static value followed by a local vacuum assignment, determines the fluctuations. For dimensionless coordinates, H has energy units, G energy-times-time-squared units, and $\hbar/\sqrt{I_0e_*}$ controls fluctuation size.

The unit-scale Gaussian has only 0.18974552836899916 probability inside the previously declared radius $\sqrt{8/3}/4$ about the chosen vertex. That control does not certify localized records. The state is a local quadratic-well ground, not the full nonlinear multiwell ground or the outcome of a preparation process. Its copies have not been physically identified with the BRAC observer. An adopted acceptance ball is not a source-selected stable quantizer or a measured efficiency.

Scientific status. The canonical calculations supply exact conditional identities and finite numerical controls, not Nature occupation. Physical observer identification, preparation, nonlinear record stability and calibrated operational resolution remain open. Standard quadratic-detector, coherent-state and coupled-Gaussian physics supply comparison methods, not evidence of a Tau-specific effect.

88 Source identification and recovery

The current observer audit separates a conditional forward construction from physical source selection. A frozen body, occupied observer incidence, preparation, complete instrument and stable resolution can determine records without selecting those inputs from Nature. Equal effects do not fix postmeasurement states, and symmetry within a chosen interaction family does not exclude other symmetric couplings. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

Dependency and claim ownership. Foundation VI and the body-readout technical companion own the finite instrument/access proof; Foundation IV and TD-0 retain physical source selection; Foundation V, the temporal synthesis and seed-body companion own the gradient/inertial distinction. This paper inherits only the stated specialization. Canonical assumptions, countermodels and calibration audits are in https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_rqm_instrument_to_record_bridge_001.md. No physical seed, Nature occupation or Tau-exclusive observation is established by this update. Existing empirical endpoint definitions and scores are unchanged.

89 Conditional source preparation and time-resolved access

The common-operator scalar–Dirac completion now supports an explicit conditional chain from interaction vertices to pair transfer and joint readout. Nonzero vertices do not select occupation. Relational coherence can be generated from an occupied number-diagonal pair even while both local mean fields vanish. In a declared resonant rotating-wave and Markov loss model, finite-window emission retains this coherence information, whereas an all-time count discards it. This is standard-compatible quantum recovery, not a Tau-specific observational residual. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

Frozen model and preparation boundary. The conditional completion uses a constant scalar kinetic metric K , a stationary potential Hessian H_* , and a common Dirac mass operator gD_* . Scalar masses obey $H_*u = M^2Ku$ and fermion masses are $g|d_i|$. The signed masses are retained in the transformed Yukawa spin factors. For the four selected pump directions, the inclusive fermionic partial widths at the stipulated $g = 1$ benchmark are 0.10031707, 0.12080077, 0.15789991, 0.14826014 in model mass units. Their ratios to scalar mass range from 0.09459 to 0.11929; these are not measured lifetimes or a parametrically narrow-resonance proof. Single-fermion emission into these pumps is kinematically closed at this point; inverse fermion–antifermion preparation has allowed on-shell momenta, but no incoming population is thereby selected.

For a canonical homogeneous mode in a declared volume, a displacement $A\sqrt{w_C}$ with zero mean velocity has minimum coherent density $n_C = M_C A^2 w_C / 2$. Positivity gives $\langle N \rangle \geq |\langle a \rangle|^2$; a displaced thermal state has the same mean and larger occupation. This is a conditional bound, not selection of a coherent initial state.

Existing interaction and relational information. On an isolated resonant pair/scalar sector let $B = |\text{empty}\rangle\langle\text{pair}|$ and $H_I = Ga^\dagger B + G^* B^\dagger a$. Choose a real positive G convention and $\theta = Gt$. An initially occupied pair probability p with no local phase coherence evolves to

$$\langle N_\phi \rangle = p \sin^2 \theta, \quad \langle a \rangle = 0, \quad \langle a^\dagger B \rangle = ip \sin \theta \cos \theta.$$

The last quantity is invariant under common total-number phase rotation. It requires occupation, not an external absolute phase reference. A single pair nevertheless cannot prepare a nonzero exact coherent oscillator state. With ideal number detection after a continuation angle δ , on the basis $|P\rangle, |S\rangle$,

$$E_\delta = \frac{1}{2}[I - \cos(2\delta)Z - \sin(2\delta)Y].$$

Three settings determine both populations and the imaginary joint coherence, but the real coherence is an exact null of this measurement family. This does not permit deleting that direction from other readouts.

Simultaneous exchange and loss. For a separately justified constant-rate Markov reservoir, distinct from the coherent pair, the no-emission operator and finite-window effect are

$$H_{\text{eff}} = \begin{pmatrix} 0 & G \\ G & -i\Gamma/2 \end{pmatrix}, \quad K_0(T) = e^{-iH_{\text{eff}}T}, \quad E_{\text{em}}(T) = I - K_0(T)^\dagger K_0(T).$$

Differentiation gives

$$E_{\text{em}}(T) = \int_0^T \Gamma K_0(t)^\dagger |S\rangle\langle S| K_0(t) dt, \quad 0 \leq E_{\text{em}} \leq I.$$

At the declared control $\Gamma/G = GT = 1$, the joint state with diagonal entries $1/2$ and off-diagonal entry $i/2$ has emission probability 0.5581543931; its dephased control has 0.3385743227. Independent quadrature and matrix exponentiation agree to below 6×10^{-16} over twenty tested windows. These are internal controls, not measurements. For $G \neq 0$ and $\Gamma > 0$, $E_{\text{em}}(\infty) = I$ on the occupied sector: all-time counting discards the distinguishing coherence. Finite timing resolution is therefore essential for this particular signal.

What is not closed. Emission is not a registered or amplified record. A collection fraction multiplies the effect; unknown efficiency and occupation remain degenerate. The normalized coherent coupling G is not the inclusive continuum width Γ , and neither fixes SI seconds. Physical preparation, mode normalization, reservoir approximation, gauge completion, detector and clock calibration, and stable Q_{OS} remain unselected. Standard quantum exchange reproduces these conditional results; no new empirical score or Tau-exclusive prediction follows.

Dependency ownership and limits. The P4 common-source assembly owns the scalar/Yukawa benchmark and finite preparation calculations; Foundation VI and the body-readout companion inherit the joint-access mathematics. Foundation IV and TD-0 retain physical source selection, and Foundation V retains clock descent. The parent remains atemporal: the evolution above is a conditional downstream 4D model, not parent memory or a derived morphogenetic time. Physical initial occupation, a gauge-complete realization, detector amplification, absolute calibration and stable observer-relative Q_{OS} remain unselected. No empirical endpoint or score is changed. The canonical derivation and reproducible numerical sources are linked at https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_core_p4_nonabelian_particle_common_source_assembly_001.md.

90 Conclusion

The full derivation ledger shows that the Higgs branch has moved from a single quartic coincidence to a structured single-package proof program. The sharp fixed chain is

$$3 + 2 \rightarrow \kappa_\tau^2 = \frac{3}{5} \rightarrow \nu_H = \frac{3}{10} \rightarrow I_4(3/10) \rightarrow \mathcal{I}_{qHu} \rightarrow y_t^{\text{parent}} = 0.9615522319206335. \quad (237)$$

The remaining blockers are now also sharp: ϵ_τ , C_A , compact cell, wall embedding, continuum regulator, top determinant, and matching. Until those are derived, this is a disciplined candidate derivation program, not a completed Higgs-sector proof.

Negative status at closure: conditional compact spectral theorem and parent-domain selection functional formulated within stated operator-domain assumptions, but exhaustive domain classification is not closed; conditional positive-spectrum determinant cancellation follows within the same-domain setup and finite-residue extraction gate is formulated, but no computed physical top finite residue; conditional anomaly/Ward and representation trace-cancellation gates formulated, but no computed parent-derived representation complex; UV/continuum admissibility criterion and microscopic completion target formulated, but no convergent parent action family; physical matching map and endpoint protocol formulated, but no numerical endpoint matching yet.